TEXAS: A proxy system model for TEX_{86} paleothermometry

Ronnakrit Rattanasriampaipong[1,2], Jessica E. Tierney[1], Felix J. Elling[3], Gordon N. Inglis[4]

[1]Department of Geosciences, The University of Arizona, Tucson, AZ, USA
[2]University Corporation for Atmospheric Research, Boulder, CO, USA
[3]Leibniz-Laboratory for Radiometric Dating and Isotope Research, Kiel University, Kiel, Germany
[4]School of Ocean and Earth Science, University of Southampton, Southampton, UK

Key Points:

- TEXAS mechanistically models TEX_{86} from archaeal physiology to sedimentary archives
- The Bayesian hierarchical model (TEXAS-Bay) incorporates nonthermal covariates (AOA ecology, nitrate), explaining 79.4% of core-top scaled RI variance
- PETM temperatures reduced 3–5°C relative to existing calibrations, improving proxy-model agreement

Corresponding author: Ronnakrit Rattanasriampaipong, ronnakritr@arizona.edu

Abstract

The TEX_{86} proxy is essential for reconstructing past ocean temperatures from archaeal membrane lipids (isoprenoid glycerol dialkyl glycerol tetraethers, or GDGTs). Existing calibrations rely on empirical relationships between observed ocean temperatures and GDGT distributions in surface marine sediments. However, reconstructing temperatures beyond the calibration range requires extrapolation, which is highly sensitive to functional form. Additionally, sedimentary TEX_{86} values are influenced by nonthermal factors—including archaeal community structure and nutrient availability—that have not been systematically incorporated into calibration frameworks.

Here, we present TEXAS (TetraEther indeX for Ammonia oxidizerS), a proxy system model that calibrates temperature against the Scaled Ring Index—a normalized (0–1) weighted average of cyclopentane rings across the six common GDGT structures. We model this relationship using a sigmoid function that captures bounded, nonlinear behavior at temperature extremes. Our hierarchical Bayesian calibration (TEXAS-Bay) uses two layers: the top layer constrains sigmoidal temperature response using laboratory cultures of ammonia-oxidizing archaea (AOA) and seawater mesocosms, extending calibration to >30°C; the bottom layer refines this using global core-top data while incorporating nonthermal covariates (AOA community structure via GDGT-2/GDGT-3 ratio and upper-ocean nitrate concentration). The full model explains 79.4% of core-top scaled RI variance (RMSE = 0.049), significantly outperforming existing calibrations.

Applications to late Quaternary glacial–interglacial cycles and the Paleocene–Eocene Thermal Maximum (PETM) demonstrate robust performance across diverse climate states. For the Quaternary tropics, TEXAS-Bay reconstructs subsurface temperatures consistent with published records; at mid-latitudes, it reduces warm biases relative to alkenone-based SST. For the PETM, TEXAS-Bay yields 4.6–7.6°C warming relative to pre-event baselines—a 3–5°C reduction compared to linear calibrations that substantially improves proxy-model agreement in high-CO_2 simulations. By explicitly accounting for archaeal ecology and nutrient effects, TEXAS provides a mechanistically grounded framework for temperature reconstruction across Mesozoic through Cenozoic climate states.

Plain Language Summary

Ocean temperatures in Earth's deep past can be reconstructed from molecular fossils preserved in marine sediments. Tiny single-celled organisms called archaea build their cell membranes using lipid molecules that contain ring-shaped structures. Warmer water causes these archaea to produce membranes with more rings. Scientists measure this relationship using an index called TEX_{86}, which works well for modern ocean conditions but becomes unreliable for climates much warmer than today. The problem is twofold: different mathematical models diverge strongly when applied beyond modern temperature ranges, and factors other than temperature—including nutrient availability and archaeal community composition—also affect the preserved chemical signals.

We developed TEXAS, a new framework that addresses these challenges. Rather than using TEX_{86} directly, TEXAS uses the Ring Index, which incorporates information from all major membrane molecules and shows better consistency between laboratory experiments and seafloor sediment data. We model the temperature relationship using an S-shaped curve that prevents unrealistic estimates at extreme temperatures. Our approach combines three data sources—laboratory cultures of archaea, controlled seawater experiments, and globally distributed marine sediments—while explicitly accounting for nutrient effects and archaeal ecology. When applied to ancient warm periods like the Paleocene–Eocene Thermal Maximum (56 million years ago), TEXAS reduces estimated warming by 3–5°C compared to previous methods, substantially improving agreement between geological temperature records and climate model simulations. This mechanistic framework provides more reliable ocean temperature reconstructions for Earth's warm climate states, from recent ice ages to ancient greenhouse intervals.

1 Introduction

The paleotemperature proxy TEX_{86} (Tetraether indeX of 86 carbons; Schouten, Hopmans, Schefuss, & Sinninghe Damsté, 2002) is commonly used to reconstruct ocean temperatures during the Mesozoic and Cenozoic Eras (Judd et al., 2022, 2024). TEX_{86} has been instrumental in advancing our understanding of Earth's climate history, demonstrating that tropical sea surface temperatures (SST) exceeded 30°C during past warm intervals (e.g., Schouten et al., 2003; Zachos et al., 2006; Frieling et al., 2017)—thereby refuting the once-hypothesized "tropical thermostat" that proposed oceanic temperatures were limited to ~30–31°C (Ramanathan & Collins, 1991). This paleothermometer is based on the strong empirical relationship between the modern sedimentary distribution of membrane-spanning lipids of planktonic marine archaea, known as glycerol dialkyl glycerol tetraethers (GDGTs), and overlying ocean temperatures (Schouten et al., 2002; Kim et al., 2010; Tierney & Tingley, 2014, 2015). Importantly, seawater mesocosms with natural populations of marine archaea confirm that this relationship remains linear and does not saturate at temperatures up to 40°C (Wuchter et al., 2004; Schouten et al., 2007), well beyond modern tropical SST maxima. The thermal sensitivity of archaeal membrane lipid biosynthesis is well-documented in both (hyper)thermophilic (De Rosa et al., 1980; Uda et al., 2001; Pearson et al., 2008) and mesophilic (Qin et al., 2015; Elling et al., 2015) cultures. These organisms employ homeoviscous adaptation, regulating membrane fluidity by varying the cyclization of GDGTs to suit ambient environmental temperatures.

Since the development of the original linear calibration between SST and TEX_{86} (Schouten et al., 2002), several calibration models have been proposed to improve statistical robustness and address regional proxy biases. These include a common logarithmic transformation of TEX_{86} (TEX^{H}_{86}; *Eq. S2* in *Table S1*; Kim et al., 2010), a spatially-varying model with a Bayesian inference framework (BAYSPAR; Tierney & Tingley, 2014, 2015), and a machine learning approach (OPTiMAL; Dunkley Jones et al., 2020) (**Fig. 1a**). However, fundamental challenges persist in temperature calibrations. *First*, existing models are exclusively derived from core-top TEX_{86} measurements calibrated to modern ocean temperatures, where the annual mean SST reaches a maximum of ~30°C. Consequently, reconstructing past climates that were warmer than today requires extrapolating the calibrations into no-analog states, resulting in temperature estimates that diverge by up to 10°C (**Fig. 1**). *Second*, laboratory culture experiments show that marine archaea adjust their membrane lipids in response to not only temperature but also other environmental factors, including growth rate (Elling et al., 2014), oxygen concentration (Qin et al., 2015), and substrate availability (Hurley et al., 2016). *Third*, while TEX_{86} serves as a (sub)surface ocean temperature proxy, marine archaea inhabit the entire water column (Karner et al., 2001). Increased export of isoprenoid lipids from deep-water archaeal clades can alter sedimentary GDGT compositions and thus bias surface-ocean signals. Although the ratio of GDGTs with two and three internal cyclopentane rings (GDGT-2/GDGT-3) has been used to detect such deep-water GDGT contributions in marine sediments (Taylor et al., 2013; Rattanasriampaipong et al., 2022), it has only recently been applied to determine archaeal export depths (e.g., van der Weijst et al., 2022).

Here, we present TEXAS (TetraEther indeX of Ammonia oxidizerS), a new proxy system model (PSM) that addresses these challenges by integrating both thermal and non-thermal factors into the modeling of GDGT distributions.

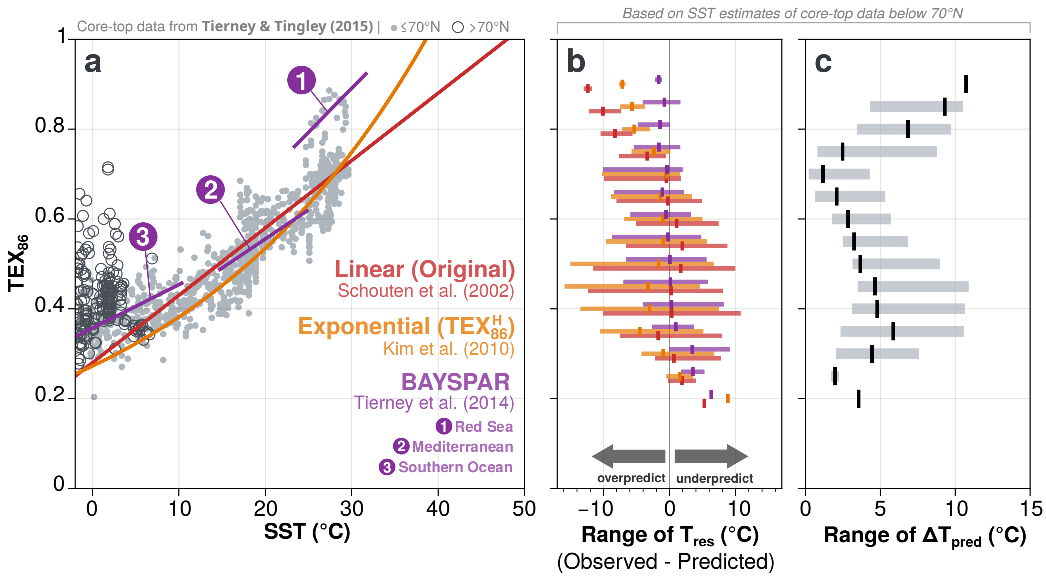

Fig. 1. Existing TEX$_{86}$-SST calibration models and SST prediction uncertainties. (**a**) A scatter plot shows the SST–TEX$_{86}$ relationship of the global core-top dataset from Tierney and Tingley (2015) overlaid with selected existing calibration models: Linear (red; Schouten et al., 2002), TEX$^H_{86}$ (orange; Kim et al., 2010), and BAYSPAR (purple; Tierney & Tingley, 2014). Dots and open circles represent data from ≤70°N and >70°N, respectively. (**b**) Range of temperature residuals (T_{res}; observed minus predicted SST) for each calibration model (colors correspond to panel (**a**) across binned TEX$_{86}$ intervals. (**c**) Range of predicted temperature differences (ΔT_{pred}) between calibration models across binned TEX$_{86}$ intervals. Gray bars show minimum-maximum range of ΔT_{pred} for the global core-top dataset.

2 TEXAS framework

TEXAS adopts a PSM framework (M. N. Evans et al., 2013) that decomposes the relationship between environmental conditions and measured proxy values into discrete mechanistic steps (**Fig. 2**). The framework comprises three submodels connecting four components:

- The `sensor model` quantifies the biological response of GDGT distributions to the environment. While traditional calibrations focus solely on temperature, the TEXAS sensor model is multivariate, formally integrating (1) ammonia-oxidizing archaea (AOA) community structure—to account for water-column depth effects (Taylor et al., 2013; Rattanasriampaipong et al., 2022)—and (2) surface ocean nitrate concentrations (Rattanasriampaipong et al., 2025).
- The `archive model` accounts for post-depositional alteration of GDGT signals, such as diagenesis of organic matter, lateral transport of archaeal lipids after deposition, or deformation of sedimentary layers.
- The `observation model` propagates analytical and chronological uncertainties.

By integrating mechanistic constraints from laboratory cultures and mesocosm experiments, TEXAS moves beyond the empirical limitations of modern core-top datasets. This PSM framework allows for more robust reconstructions of past warm intervals that lack modern thermal analogs. The modular framework presented here can be readily extended to incorporate additional factors as they are identified through future experimental and observational studies.

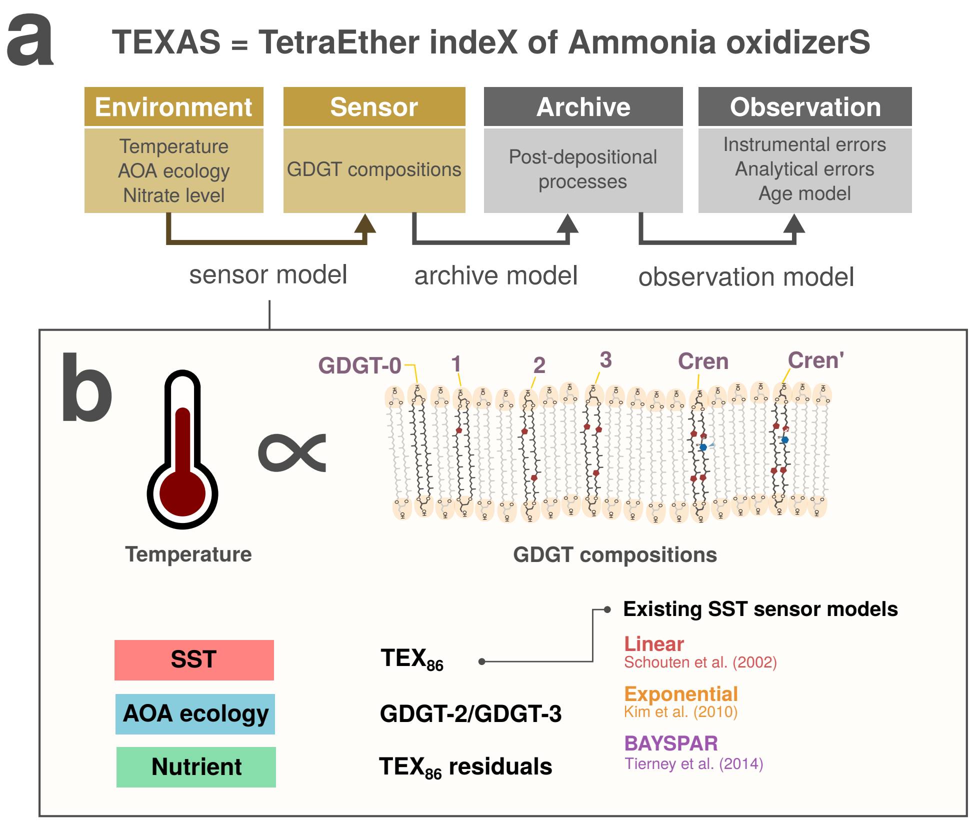

Fig. 2. The TEXAS proxy system modeling framework. (**a**) TEXAS comprises three submodels (*sensor*, *archive*, and *observation*) connecting four components following M. N. Evans et al. (2013): *environment* (ambient conditions), *sensor* (biological response), *archive* (preservation), and *observation* (measurement). (**b**) The *sensor* model links environmental factors to measurable GDGT indices: temperature controls TEX_{86} (e.g., Schouten et al., 2002; Kim et al., 2010; Tierney & Tingley, 2014), AOA ecology controls the GDGT-2/GDGT-3 ratio (cf. Taylor et al., 2013; Rattanasriampaipong et al., 2022), and nutrient availability influences TEX_{86} residuals (observed minus predicted TEX_{86}) (cf. Rattanasriampaipong et al., 2025).

3 An extended GDGT database for temperature calibration

We used the extended global core-top GDGT dataset recently published in Rattanasriampaipong et al. (2025), which contains 2,084 unique TEX_{86} measurements from core-top sediments spanning global ocean basins (**Fig. 3a**; Schouten et al., 2002; Kim, Schouten, Hopmans, Donner, & Sinninghe Damsté, 2008; Trommer et al., 2009; Seki et al., 2009; Kim et al., 2010; Leider, Hinrichs, Mollenhauer, & Versteegh, 2010; Castañeda et al., 2010; Shevenell, Ingalls, Domack, & Kelly, 2011; Y. Wei et al., 2011; Ho, Yamamoto, Mollenhauer, & Minagawa, 2011; Chazen, 2011; Richey, Hollander, Flower, & Eglinton, 2011; Verleye, 2011; Jia, Zhang, Chen, Peng, & Zhang, 2012; Fallet et al., 2012; Hu, Meyers, Chen, Peng, & Yang, 2012; Wu, Tan, Zhou, Yang, & Xu, 2012; Nieto-Moreno et al., 2013; Smith et al., 2013; Bale, Villanueva, Hopmans, Schouten, & Sinninghe Damsté, 2013; W. Chen, Mohtadi, Schefuß, & Mollenhauer, 2014; Tierney & Tingley, 2014; Hernández-Sánchez, Woodward, Taylor, Henderson, & Pancost, 2014; Lengger, Hopmans, Sinninghe Damsté, & Schouten, 2014; Zell et al., 2014; Lü et al., 2014; Ho et al., 2014; X.-L. Liu, Zhu, Wakeham, & Hinrichs, 2014; Park et al., 2014; Seki et al., 2014; Zhou, Hu, Spiro, Peng, & Tang, 2014; Kaiser et al., 2014; Tierney & Tingley, 2015; Tierney, Ummenhofer, & DeMenocal, 2015; Kaiser et al., 2015; Kim, Villanueva, Zell, & Sinninghe Damsté, 2016; Pan et al., 2016; Richey

& Tierney, 2016; Kusch, Rethemeyer, Hopmans, Wacker, & Mollenhauer, 2016; Sinninghe Damsté, 2016; Jaeschke et al., 2017; J. Chen et al., 2018; Ceccopieri, Carreira, Wagener, Hefter, & Mollenhauer, 2018; ?, ?; Lo et al., 2018; Yang et al., 2018; Harning et al., 2019; B. Wei et al., 2020; Lamping et al., 2021; Sinninghe Damsté, Warden, Berg, Jürgens, & Moros, 2022; Hagemann et al., 2023; Harning, Holman, Woelders, Jennings, & Sepúlveda, 2023; Varma, Hopmans, et al., 2024; Rattanasriampaipong et al., 2025).

In addition to this core-top compilation, we collated GDGT measurements from seawater mesocosms ($n = 21$; Wuchter et al., 2004; Schouten et al., 2007) and laboratory culture experiments with known strains of ammonia-oxidizing archaea (AOA) from marine ($n = 42$; Sinninghe Damsté, Schouten, Hopmans, van Duin, & Geenevasen, 2002; Pitcher et al., 2011; Qin et al., 2015; Elling et al., 2015, 2017; Varma, Villanueva, et al., 2024), soil ($n = 5$; Jung et al., 2011; Sinninghe Damsté, Ossebaar, Schouten, & Verschuren,

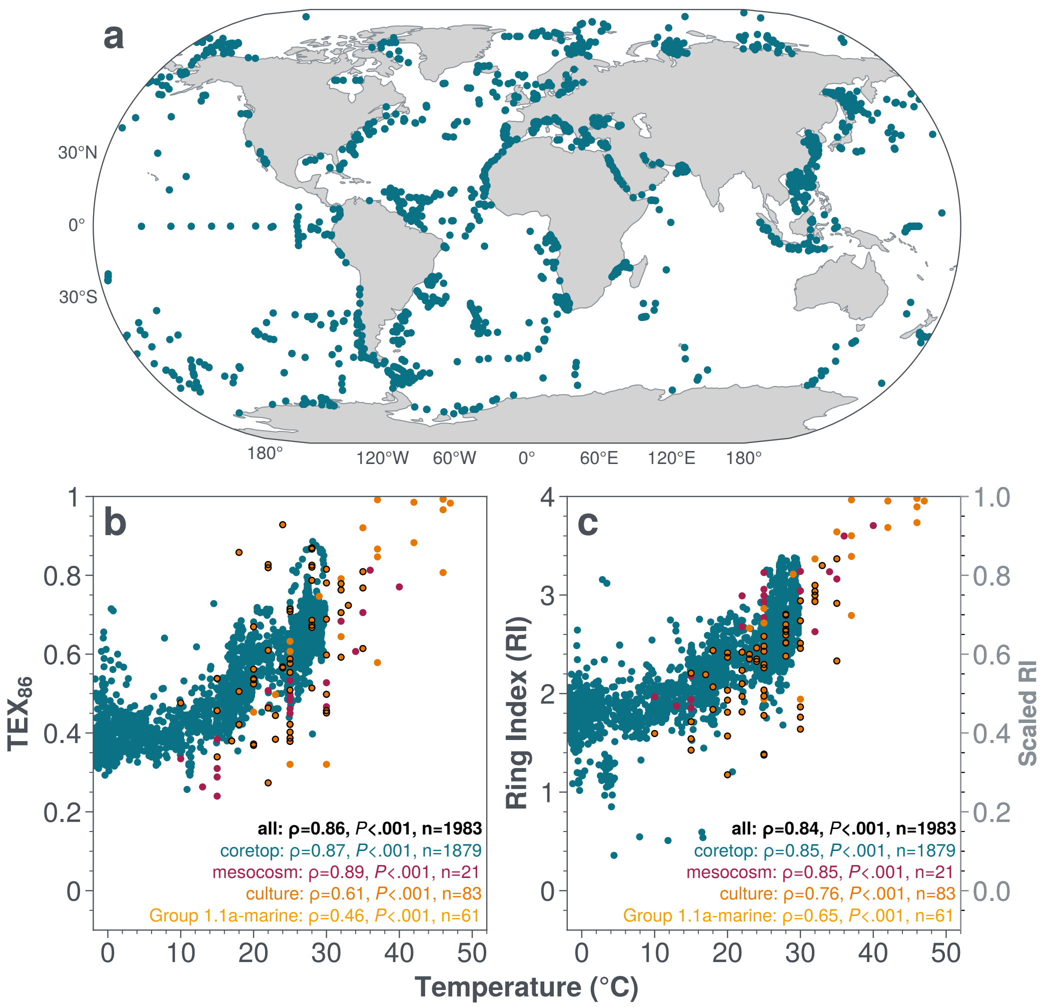

Fig. 3. Extended calibration dataset. (**a**) Locations of the extended global core-top TEX_{86} dataset from Rattanasriampaipong et al. (2025). (**b**) and (**c**) shows responses of TEX_{86} and Ring Index (RI) over annual mean sea surface temperatures derived from the World Ocean Atlas 2023 0.25-degree grid temperature product (Locarnini et al., 2024). Low abundance data (<0.001 in any GDGT fractions) are excluded. Data are color-coded by data types: core-top (dark cyan), culture (orange), mesocosm (magenta). Cultures from Marine Group 1.1a strains are highlighted with black marker edges. Spearman's correlation coefficient (ρ), associated P-values, number of observations (n) of all data with nearest-matched temperatures of each data type are also reported.

2012; Jung et al., 2014, 2016), and hot spring ($n = 8$; Pitcher et al., 2010; Elling et al., 2017; Bale et al., 2019) environments. We also report new GDGT measurements of archaeal cultures from marine ($n = 22$) and soil ($n = 9$) environments [Felix's unpublished data; Felix please help revise the text as appropriate]. Enrichment and culturing conditions for all culture experiments are described in the original studies ***Table S2***.

The combined dataset contains 2,191 data points representing three types (core-top, culture, mesocosm). We applied two quality control filters: (1) removal of 59 core-top entries lacking individual GDGT fraction abundances, and (2) exclusion of 31 entries (28 core-top, 3 culture) where any isoprenoid GDGT fraction had a relative abundance below 0.001. We also excluded one core-top data from Kusch et al. (2016) as the original work reported that the sample is subject to analytical errors. This yielded a final dataset of 2,100 entries for the subsequent analysis for our calibration model development.

Following Rattanasriampaipong et al. (2025), we matched each core-top GDGT measurement to environmental parameters from the nearest gridpoint in global ocean datasets: (1) sea surface temperature (SST) and thermocline-integrated temperature (thermo-T) from the World Ocean Atlas 2023 (WOA23) 0.25-degree gridded product (Locarnini et al., 2024), and (2) thermocline-integrated nitrate concentration (nitrate or NO_3^-) from the Copernicus Marine Environment Monitoring Service (CMEMS) 0.25-degree global ocean biogeochemistry reanalysis (Perruche, 2018).

4 Ring Index as a predictand

GDGTs are macrocyclic lipids comprising two C_40 biphytane backbones linked by ether groups. In marine settings, sedimentary GDGT pools commonly contain isoprenoid GDGTs with up to three internal cyclopentane rings (GDGT-0 to -3), along with two structurally unique compounds—crenarchaeol (cren) and its stereoisomer (cren')—that each contain four cyclopentane rings and one cyclohexane ring (**Fig. 2b**). Crenarchaeol is considered a biomarker specific to marine AOA belonging to the class *Nitrososphaeria* as it has not been observed in other archaeal lineages (Sinninghe Damsté et al., 2002; Elling et al., 2025).

TEX_{86} is the ratio of a subset of GDGTs that exclude GDGT-0 and Crenarchaeol (*Eq. S1* in *Table S1*) and was originally selected as the ratio for temperature calibration because its linear correlation with SST produced the highest coefficient of determination (r^2) in the initial dataset ($r^2 = 0.92$; $n = 42$; Schouten et al., 2002). The GDGT-0 fraction was intentionally excluded from the TEX_{86} to minimize compositional biases of GDGTs from non-marine AOA sources (Schouten et al., 2002, 2013; Elling et al., 2025). This view was supported by the lower r^2 values ($r^2 = 0.63$) in the original dataset when correlating SST with the weighted average ring numbers of all isoprenoid GDGT fractions. However, this empirical view has rarely been updated with expanded core-top compilations, and subsequent calibration studies have shown that the relationship between TEX_{86} and SST is not best explained by simple linear regression (Schouten et al., 2013; Inglis & Tierney, 2020; Elling et al., 2025). Moreover, the exclusion of GDGT-0 may not be optimal, as this GDGT fraction exhibits the strongest response to temperature changes when individual GDGT fractions are correlated with temperature (*Fig. S1*).

We therefore re-evaluate the temperature response of the weighted average number of cyclopentane rings using the Ring Index (RI) (Zhang et al., 2016). Although several formulations to calculate average ring numbers exist, we adopt Zhang et al. (2016)'s formulation (*Eq. S3* in *Table S1*) due to its established use in the paleoceanographic community as a screening metric in TEX_{86} studies. To facilitate direct comparison with TEX_{86} (which ranges from 0 to 1), we normalize RI by its theoretical maximum of 4 to a 0–1 range (*Eq. S4* in *Table S1*), hereafter referred to as scaled RI. Using the complete unscreened dataset, comparison of TEX_{86} and RI reveals comparable Spearman's rank correlation coefficients (ρ) with temperature for the full dataset ($\rho_{\mathrm{TEX}_{86}} = 0.86$ vs. $\rho_{\mathrm{RI}} = 0.84$), extended core-top data (0.87 vs. 0.85), and mesocosm experiments (0.89 vs. 0.85) (**Fig. 3b,c**). Notably, in archaeal cultures RI has a much stronger temperature response thanTEX_{86} (0.76 vs. 0.61) and this is particularly the case for the Marine Group 1.1a subset (0.65 vs. 0.46) (cf. Elling et al., 2025). This enhanced sensitivity across diverse data types reinforces the observation that GDGT-0, which is excluded from TEX_{86}, is a vital component of the thermal biosynthesis of archaeal membrane lipids (*Fig. S1*).

In addition, visual inspection demonstrates that culture and mesocosm data converge more closely with the core-top dataset in RI-temperature space (**Fig. 3c**) than TEX_{86}-temperature space (**Fig. 3b**). This reduced offset between experimental and sedimentary datasets is critical, as we can now extend calibrations beyond modern ocean SST ranges using laboratory constraints from culture and mesocosm experiments with greater confidence. Thus, we select RI as the predictand in our subsequent analysis.

5 Data Screening

Sedimentary GDGT assemblages can receive contributions from multiple archaeal sources beyond planktonic marine AOA, potentially biasing temperature signals from the upper ocean. Traditional screening indices—including the Branched versus Isoprenoid Tetraether index (BIT; Hopmans et al., 2004), Methane Index (MI; Zhang et al., 2011), %GDGT-0 (Blaga et al., 2009; Sinninghe Damsté et al., 2012; Inglis et al., 2015), and ΔRing Index (ΔRI; Zhang et al., 2016)—were developed to identify samples unsuitable for temperature reconstruction. However, these metrics rely on arbitrary thresholds, are inconsistently applied, and can systematically exclude valid marine samples (see *Supplement Text S1* for detailed discussion).

Of particular concern, the widely used ΔRI metric (Zhang et al., 2016)—which flags samples deviating from a quadratic TEX_{86}–RI relationship ($|\Delta\mathrm{RI}| > 0.3$)—excludes 22% of our core-top dataset and 62% of Marine Group 1.1a cultures (*Fig. S2 and S3*). Critically, excluded samples are disproportionately characterized by low GDGT-2/GDGT-3 ratios ($\leq$5), a fingerprint of shallow-water AOA (Taylor et al., 2013; Rattanasriampaipong et al., 2022), suggesting systematic removal of valid information for temperature calibration (**Fig. 4c**, *S5e–g*). These limitations motivated development of a data-driven screening approach.

5.1 Mahalanobis distance ellipsoid: a new approach for GDGT data screening

Following the agnostic, data-informed logic of the ΔRI metric, we model the GDGT distribution of "normal" marine sediments as a Gaussian covariance structure within the TEX_{86}–RI space. Using the `scipy.spatial.distance.mahalanobis` function, we calculated Mahalanobis distances (D_M) between each data point and the reference distribution's centroid. Unlike the $\mathrm{D}_{\mathrm{nearest}}$ metric used in the OPTiMAL calibration framework (Dunkley Jones et al., 2020)—which assumes a spherical distribution via Euclidean distance across six GDGT fractions—the D_M accounts for the specific ellipsoid geometry and correlation between TEX_{86} and RI.

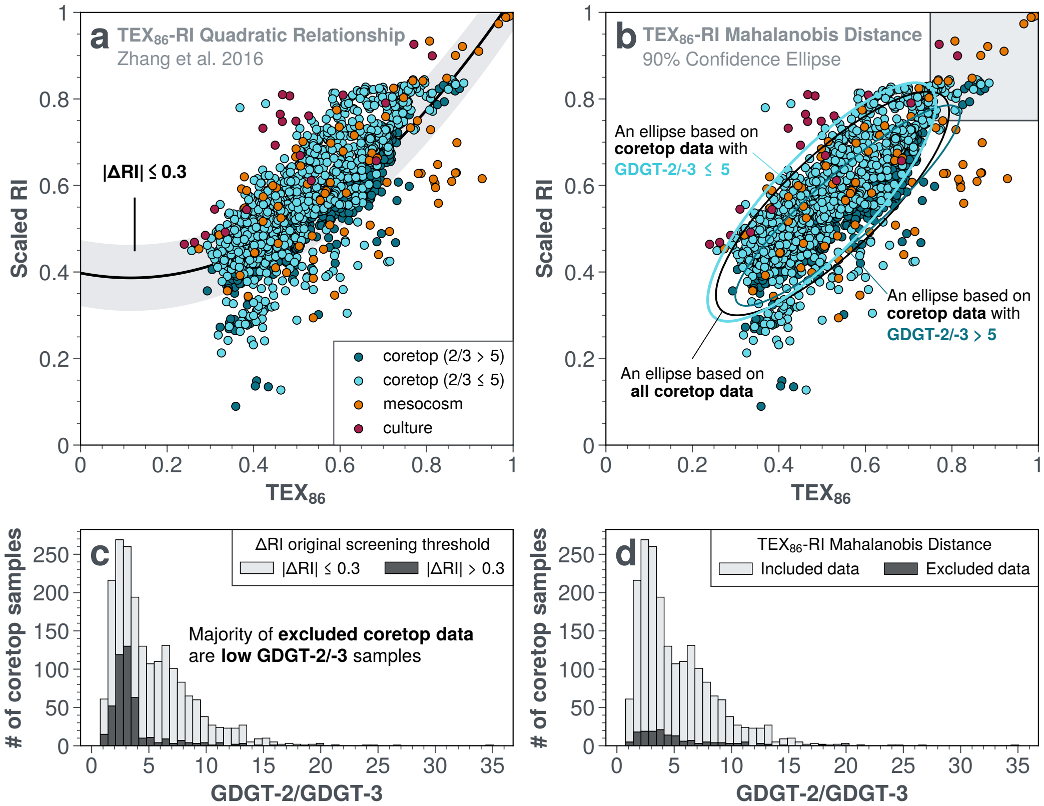

Fig. 4. Comparison of $\triangle$RI and Mahalanobis distance screening approaches in TEX$_{86}$–RI space. (**a**) The quadratic TEX$_{86}$–RI relationship from Zhang et al. (2016) with the original $|\triangle\text{RI}| \leq 0.3$ screening threshold (gray band). (**b**) Mahalanobis distance screening based on 90% confidence ellipses calculated from three reference distributions: all core-top data (black ellipse), core-top data with GDGT-2/GDGT-3 ≤ 5 (upper cyan ellipse), and core-top data with GDGT-2/GDGT-3 > 5 (lower dark cyan ellipse). Data with TEX$_{86}$ and scaled RI values above 0.75 are preserved in the calibration dataset. (**c, d**) Histograms show the GDGT-2/GDGT-3 distribution of the extended core-top dataset (n = 1,996) with samples classified as included (light gray bars) or excluded (dark gray bars) by (**c**) the traditional $|\triangle\text{RI}| \leq 0.3$ threshold and (**d**) the Mahalanobis distance approach using core-top data with GDGT-2/GDGT-3 ≤ 5 as a reference cluster.

The reference distribution was defined using core-top data with GDGT-2/GDGT-3 ≤ 5, identifying samples where sedimentary GDGTs in these samples are primarily derived from shallow-water AOA clades (cf. Taylor et al., 2013; Rattanasriampaipong et al., 2022). We established a screening threshold based on the χ^2 distribution (2 degrees of freedom) at a 90% confidence level; points exceeding this distance threshold from the centroid are flagged as outliers. This confidence level was selected via sensitivity testing to maximize the correlation between TEX$_{86}$ and RI while maintaining a robust calibration dataset (*Fig. S4*).

To preserve critical constraints at the warm end of the calibration range, we retain all samples with TEX$_{86} \geq 0.75$ and scaled RI ≥ 0.75, regardless of their D_M (**Fig. 4b**). Under this framework, outliers are defined not by their misfit to a specific regression model, but by their multivariate distance from the centroid of the primary TEX$_{86}$–RI cluster. This allows the dataset to autonomously identify GDGT assemblages inconsistent with typical thermal responses. Importantly, this approach avoids the systematic exclusion of samples with low GDGT-2/GDGT-3—a notable limitation of the original ΔRI screening (**Fig. 4c vs. 4d**). Sensitivity tests reveal that repeating the D_M calculation using all six GDGT fractions ($D_{\text{M-allGDGTs}}$) results in the near-total exclusion of laboratory culture and mesocosm data (*Fig. S5*). Similarly, applying the OPTiMAL D_{nearest} metric with a 0.5 distance threshold (Dunkley Jones et al., 2020) to our extended core-top dataset selectively excludes samples with low GDGT-2/GDGT-3 ratios (*Fig. S5b,f*). This likely stems from the fact that the current OPTiMAL D_{nearest} calculation is based on the reference core-top collection of Tierney and Tingley (2015), which lacks the broader range of GDGT distributions present in our expanded global compilation.

After D_M screening, we calculated median values of the environmental predictors (SST, nitrate, etc) for all core-top data falling within each gridpoint to mitigate the impact of spatial clustering on regression parameters (cf. Tierney et al., 2019; Rattanasriampaipong et al., 2025). After gridding, the dataset comprises 1,540 observations: 1,477 gridded core-top entries, 8 mesocosm experiments, and 55 culture experiments (including 41 from Marine Group 1.1a). Of the 1,477 gridded core-top entries, 1,412 entries have nearest-matched temperatures (SST and thermo-T) and 1,350 have both nearest-matched temperature and nitrate values from global ocean products. The gridded core-top with both T and nitrate data is then used for all subsequent analyses, hereafter called the *global calibration dataset*.

6 Model Development: Functional Forms and Environmental Predictors

6.1 Temperature response of the global calibration dataset

Temperature exerts a robust, nonlinear control on scaled RI across the global calibration dataset. The global calibration dataset exhibits a strong correlation with both SST ($\rho = 0.85$) and thermo-T ($\rho = 0.85$). D_M screening substantially improves RI-temperature correlations in the mesocosm (from $\rho = 0.89$ to 0.93) and full culture (from $\rho = 0.61$ to 0.79) datasets (**Fig. 3c vs. 5c**). In contrast, the Marine Group 1.1a subset shows a slightly weaker positive correlation after screening ($\rho = 0.65$ to 0.62). For the core-top dataset, the correlation remains unchanged relative to SST ($\rho = 0.85$) but strengthens marginally when compared against thermocline-integrated temperature ($\rho = 0.86$; thermo-T), yielding a modest improvement in overall calibration performance.

Previous calibrations of individual datasets report that temperature explains at least 78% of data variance in univariate regressions (cf. Schouten et al., 2002; Wuchter et al., 2004; Schouten et al., 2007; Kim et al., 2010, ; see *Table S2*), whether fitting TEX_{86} as a function of temperature or vice versa, using either linear or common logarithmic ($\log_{10}$) functional forms. However, applying these same functional forms to our the global calibration dataset (with expanded core-top observations) yields substantially lower r^2 values (0.66–0.69), indicating that neither function adequately captures the full data distribution in RI-temperature space. Visual inspection reveals asymptotic behavior at both temperature extremes: culture and mesocosm data approach an upper asymptote as temperatures exceed 40°C, while core-top data approach a lower asymptote below 10°C (**Fig. 5c–d**). This nonlinear behavior is confirmed by examining local regression slopes within 10°C temperature intervals (**Fig. 5e–f**), which show maximum sensitivity around 30–40°C and diminished slopes at both temperature extremes.

To capture these nonlinearities, we model the relationship between RI and temperature using a generalized logistic (sigmoid) function (also known as Richards's curve; Richards, 1959), a flexible S-shaped curve that is widely used for bounded biological responses and capable of capturing asymptotic behavior at temperature extremes:

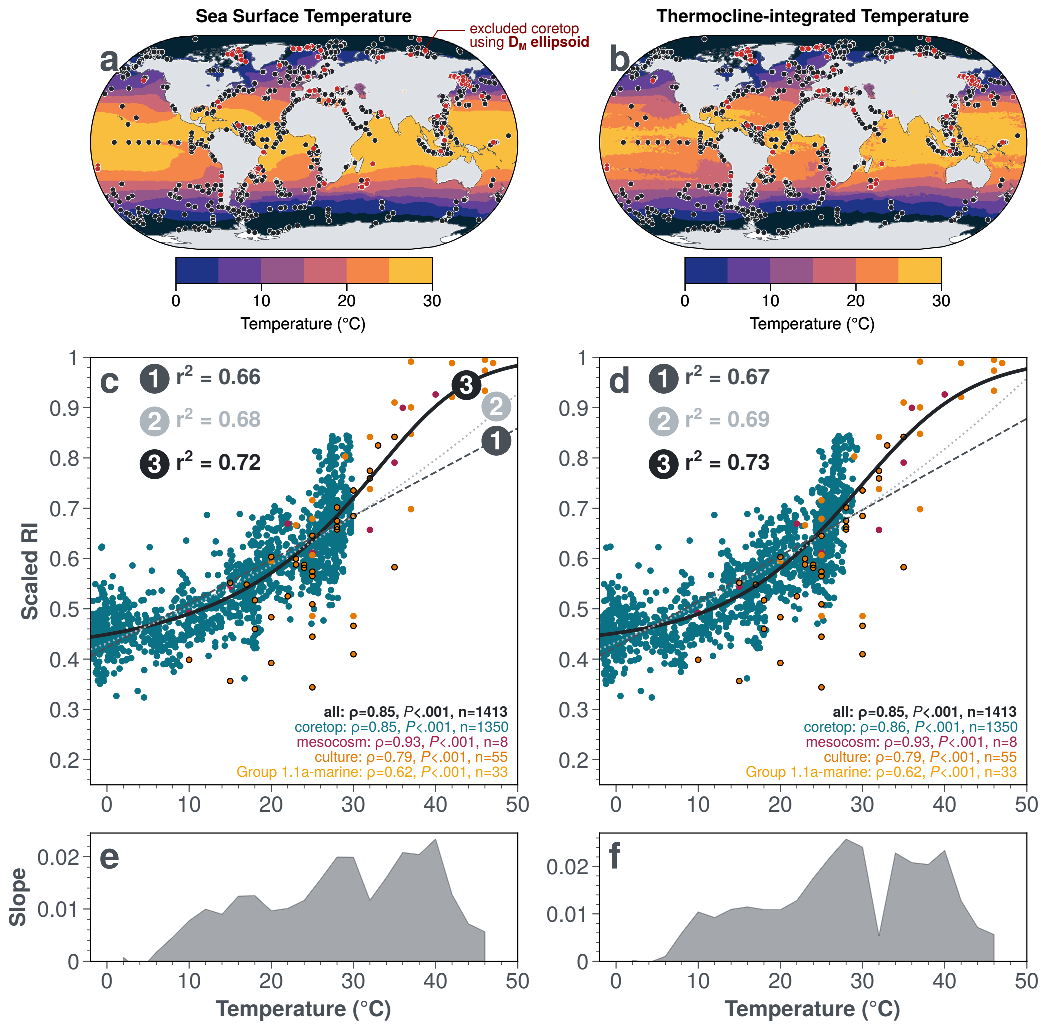

Fig. 5. Global calibration data distribution and RI–temperature functional forms. (**a**, **b**) core-top sample locations overlaid on annual mean temperature fields from the 0.25° × 0.25° WOA23 data product: black circles indicate samples retained after D_M screening; red circles indicate excluded samples. Panel **a** shows sea surface temperature (SST); Panel **b** shows thermocline-integrated temperature (thermo-T). (**c**, **d**) Scaled RI versus temperature for core-top (teal), mesocosm (maroon), and culture (orange) data. Marine Group 1.1a cultures are highlighted with black edges. Three candidate functional forms are shown: (1) linear (dashed line), (2) common logarithmic ($\log_{10}$; dotted line), and (3) generalized logistic (solid line). Core-top data in Panel **c** are plotted against SST; in Panel **d** against thermo-T. Coefficient of determination (r^2) quantifies the fit of each functional form. (**e**, **f**) Temperature sensitivity of scaled RI to temperature, calculated as linear regression slopes within 10°C sliding windows centered at 2°C intervals from 2 to 46°C.

$$\text{Scaled RI} = b + \frac{a - b}{(C + Q \cdot \exp[-k(T - T_0)])^{1/\nu}} \tag{1}$$

where T represents temperature (°C), and the model parameters are:

- a: Upper asymptote, representing the maximum RI value.
- b: Lower asymptote, representing the minimum RI value.
- T_0: Inflection point temperature, where the RI response transitions most rapidly.
- k: Curve growth rate parameter, controlling the steepness of the curve.
- ν: Shape parameter, allowing for asymmetric curves.

- Q: Shift parameter determining the initial curve position relative to T_0.
- C: Scaling constant, conventionally set to 1.

We fix the upper asymptote at $a = 1$, reflecting the theoretical maximum of scaled RI (0 to 1 range), and set $C = 1$ following standard convention, leaving five free parameters to estimate from the data (cf. Gottschalk & Dunn, 2005; Cumberland et al., 2015). This S-shaped functional form offers two key advantages over traditional linear or logarithmic models: (1) it naturally accommodates the bounded nature of the index ($0 \leq$ scaled RI ≤ 1), precluding physically impossible predictions at environmental extremes, and (2) the flexible parameterization allows the data to determine the degree of asymmetry (via ν) rather than imposing a specific functional geometry *a priori*. Supplementary figure *S6* illustrates how systematic variations in each free parameter (T_0, b, k, ν, and Q) control distinct aspects of the sigmoid curve geometry. The fitted parameters of the sigmoid function from this global calibration dataset are:

- $\theta_{\text{global-SST}}$: $T_0 = 33.951°\text{C}$, $b = 0.408$, $k = 0.209$, $\nu = 3.010$, $Q = 2.545$
- $\theta_{\text{global-thermoT}}$: $T_0 = 28.605°\text{C}$, $b = 0.428$, $k = 0.156$, $\nu = 1.608$, $Q = 1.918$

Regression analysis of RI with temperature using the sigmoidal function shows a substantial increase in r^2 values relative to the linear and logarithmic forms, with 0.72 and 0.73 for SST and thermo-T, respectively (**Fig. 5**). Importantly, the residual distribution of scaled RI (scaled $\text{RI}_{\text{res,T}}$: observed - predicted) yields lower root mean squared errors (RMSE) and shows no systematic structure across the entire temperature range (*Fig. S7a*, *S8a,d*). In contrast, scaled $\text{RI}_{\text{res,T}}$ from linear and common logarithmic regressions positively correlate with temperature for culture and mesocosm data (*Fig. S8b,c,e,f*), indicating systematic model misfit at warm temperature extremes. The lack of residual structure in the sigmoid model confirms its superiority in capturing the nonlinear temperature response and validates its use for subsequent analysis of environmental covariates.

6.2 Secondary Environmental Effects on the Gridded Core-top Dataset

After fitting the temperature response of RI with a sigmoid, we evaluated two additional environmental influences on scaled RI in the core-top data—AOA ecology (proxied by GDGT-2/GDGT-3) and nutrient stress (proxied by thermocline-integrated nitrate). To assess their individual influences, we performed univariate linear correlation analyses of the temperature residuals (scaled $\text{RI}_{\text{res,T}}$) against both predictors (*Fig. S7*).

Both variables exhibit significant negative correlations with the residuals: $\rho = -0.398$ ($P < .001$) for GDGT-2/GDGT-3 (*Fig. S7b*) and $\rho = -0.364$ ($P < .001$) for NO_3 (*Fig. S7c*, teal points). Building on the findings of Rattanasriampaipong et al. (2025), we applied a low-NO_3 threshold to isolate the core-top data most susceptible to "nutrient stress." We found that a threshold of NO3 < 1.77 μmol/L maximizes the negative correlation between scaled RIres,T and nitrate concentrations. Samples exceeding this threshold were excluded from the correlation analysis, as the nutrient effect is physiologically expected only in nitrate-depleted waters (Rattanasriampaipong et al., 2025).

Given these significant univariate responses, we constructed a multivariate forward model for core-top scaled RI:

$$
\begin{aligned}
\textbf{Scaled RI} = & f_{\text{sigmoid}}(\mathbf{T}; \theta_{\text{global}}) \\
& + \beta_{\text{G}_{2/3}} \cdot \textbf{GDGT-2/GDGT-3} \\
& + \beta_{\text{NO}_3} \cdot \log_{10}(\mathbf{NO_3}) \cdot \mathbb{I}_{(\mathbf{NO_3}<1.77)} + \epsilon
\end{aligned} \tag{2}
$$

where:

- f_{sigmoid} is the generalized logistic temperature function with parameters fitted from the global calibration dataset.
- $\beta_{\text{G}_{2/3}}$ and β_{NO_3} are the regression coefficients for the secondary predictors.
- $\mathbb{I}_{(\mathbf{NO_3}<1.77)}$ is an indicator function that equals 1 when nitrate is below the threshold (1.77 μmol/L) and 0 otherwise.
- ϵ represents the residual error term with mean zero and variance σ^2.

To assess the relative influence of the two nonthermal parameters, we performed a simultaneous multivariate regression on the gridded core-top dataset. The combined frequentist model (T + GDGT-2/GDGT-3 + NO_3) explains 77.7% of the variance in core-top scaled RI (**Table 1A**), a significant 4.1% improvement over temperature alone ($\Delta r^2 = 0.041$).

Variance partitioning identifies a notable suppression effect between GDGT-2/GDGT-3 and nitrate. Although adding GDGT-2/GDGT-3 to the temperature-only model individually yields a marginal improvement ($\Delta r^2 = 0.006$), its *unique contribution* within the full multivariate model is nearly four times larger ($\Delta r^2 = 0.022$; **Table 1B**). This amplification is particularly striking because the predictors are statistically independent ($\rho = -0.014$; *Fig. S7d*), indicating they capture distinct, orthogonal nonthermal biases that exert opposing effects on the scaled Ring Index. High GDGT-2/GDGT-3 ratios drive scaled RI (or TEX_{86}) **lower** through enhanced contributions from deep-water AOA clades (Taylor et al., 2013; Rattanasriampaipong et al., 2022), while low nitrate drives scaled RI **higher** via physiological responses to nutrient stress (Rattanasriampaipong et al., 2025).

We hypothesize that, in bivariate models (Cases 3–4 in **Table 1**), the unmodeled process introduces noise that attenuates the regression slope of the included variable (attenuation bias), a classic suppression effect in multivariate regression. Simultaneous estimation in the multivariate model eliminates this interference: by explicitly modeling the "warming" bias of nitrate stress, the model can more accurately quantify the "cooling" bias of deep-water export, and vice versa. Consequently, both regression coefficients nearly double in magnitude compared to their bivariate fits (**Table 1A**). Partial F-statistics confirm that while both predictors are significant, nitrate acts as the dominant nonthermal control ($F = 108.83$ vs. 34.65).

The multivariate coefficients quantify the physical impact of these variables on temperature reconstructions. The negative GDGT-2/GDGT-3 coefficient ($\beta_{\text{G}_{2/3}}$ = -0.0026) indicates that samples with GDGT-2/GDGT-3 > 15 experience a "cooling" bias of approximately 0.04 scaled RI units due to deep-water GDGT contributions. This magnitude exceeds the reported interlaboratory analytical precision of TEX_{86} (≈0.03 units; Schouten et al., 2009, 2013). This demonstrates that the ecological signal is distinguishable from measurement noise, though further quantification of analytical error specifically for scaled RI is warranted.

Similarly, the negative nitrate coefficient (β_{NO_3} = -0.0351) suggests that AOA in nutrient-depleted environments produce higher scaled RI values at a given temperature (cf. Rattanasriampaipong et al., 2025). This likely reflects a physiological response to energy limitation (Valentine, 2007), wherein AOA increase GDGT cyclization to optimize membrane fluidity and stability when metabolic resources—either electron donors (substrate availability; Elling et al., 2014; Hurley et al., 2016; T. W. Evans et al., 2018) or acceptors (oxygen; Qin et al., 2015)—are scarce. While these frequentist results establish the functional form and statistical significance of nonthermal effects, we integrate these constraints into a Bayesian hierarchical model (**Section 7**) to formally propagate calibration uncertainty and provide probabilistic paleotemperature reconstructions.

Table 1: Multivariate calibration model performance for core-top dataset with thermo-T

A: Regression Model Metrics

Case	Model Predictors	$\beta_{G_{2/3}}$	β_{NO_3}	r^2	Δr^2	RMSE	F-stat	*P*-val
1	T (n = 1,413; global)	–	–	0.734	–	0.059	–	–
2	T (n = 1,350; core-top)	–	–	0.736	–	0.055	–	–
3	T + GDGT-2/GDGT-3	-0.0013	–	0.742	0.006	0.054	34.65	$<.001$
4	T + NO_3†	–	-0.0243	0.755	0.019	0.053	108.83	$<.001$
5	**T + GDGT-2/GDGT-3 + NO_3†**	**-0.0026**	**-0.0351**	**0.777**	**0.041**	**0.051**	**126.00**	**$<.001$**

B: Variance Partitioning (Unique Contribution)

Component	Unique Δr^2	Interpretation
GDGT-2/GDGT-3	0.022 (2.2%)	**(5) minus (4)**: Improvement over T + NO_3 model
NO_3	0.035 (3.5%)	**(5) minus (3)**: Improvement over T + GDGT-2/GDGT-3 model

Notes:

- Temperature parameters (θ_T: ν, b, k, Q, T_0) are fixed to globally-calibrated thermo-T for all cases.
- Δr^2 in Part A shows an improved r^2 relative to **Case 2**.
- β coefficients are **re-optimized** for each case.
- †Nitrate coefficient applies only below threshold of 1.77 μmol/L.

7 TEXAS-Bay: Bayesian Calibrations for the Scaled Ring Index

Following the BAYMAG framework (Tierney et al., 2019), we developed a two-layer hierarchical Bayesian model for scaled RI. The top layer establishes the fundamental temperature-RI relationship using culture and mesocosm data, while the bottom layer refines this relationship by incorporating environmental covariates from the extended core-top dataset. Parameters were estimated via Bayesian inference using Markov chain Monte Carlo (MCMC) sampling (Gelman et al., 1995) implemented in Stan, version 2.36 (Carpenter et al., 2017).

7.1 Model Design

7.1.1 Top Layer: Culture-Mesocosm Temperature Model

The top layer (**Eq. 3**) defines the scaled RI response to temperature using laboratory culture and mesocosm datasets. This regression follows the sigmoidal form established in **Eq. 1**:

$$\begin{gathered}\textbf{Scaled RI}_{\text{culmeso}} = f_{\text{sigmoid}}(T; \theta_{\text{culmeso}}) + \epsilon_{\text{culmeso}} \\ \epsilon_{\text{culmeso}} \sim \mathcal{N}(0, \sigma^2_{\text{culmeso}})\end{gathered} \tag{3}$$

The complete parameterization of the sigmoidal component is:

$$f_{\text{sigmoid}}(T;\theta_{\text{culmeso}}) = b_{\text{culmeso}} + \frac{1 - b_{\text{culmeso}}}{(1 + Q_{\text{culmeso}} \cdot e^{-k_{\text{culmeso}}(T - T_{0,\text{culmeso}})})^{1/\nu_{\text{culmeso}}}} \tag{4}$$

The hyperparameters for the culture-mesocosm temperature response are distributed as:

$$\theta_{T_{\text{culmeso}}} \sim \mathcal{N}(\mu_{\theta_T}, \sigma^2_{\theta_T}) \tag{5}$$

The posterior distributions of these parameters ($\theta_{T_{\text{culmeso}}}$) serve as informative priors for the temperature component in the bottom layer (**Fig. 6b**).

To validate the top layer, we fit the model independently using only culture and mesocosm data. This layer achieves $r^2 = 0.726$ and RMSE $= 0.094$ (based on the median ensemble prediction of 4 chains $\times$ 1000 draws), demonstrating strong predictive skill consistent with our frequentist global calibrations (**Section 6.1**). We also evaluated temperature sensitivity by calculating the first derivative (slope) of the median posterior prediction. Unlike linear models, the sigmoidal function captures non-constant sensitivity: the maximum slope (0.025) occurs at 36.7°C and declines asymmetrically toward both temperature extremes (**Fig. 6c**). For comparison, linear regression of the combined culture-mesocosm data yields a constant slope of 0.017, while independent fits to mesocosm and culture data yield slopes of 0.014 and 0.018, respectively (**Fig. 6c**). These values align with previously reported mesocosm sensitivities (0.015–0.017; *Table S2*; cf. Wuchter et al., 2004; Schouten et al., 2007), indicating that scaled RI and TEX_{86} maintain comparable temperature responses despite their different mathematical structures.

7.1.2 Bottom Layer: Core-top Multivariate Model

The bottom layer of the TEXAS-Bay incorporates environmental and ecological variables, describing scaled RI as a function of temperature, AOA ecology (via GDGT-2/GDGT-3), and nitrate concentrations using gridded core-top data as expressed in **Eq. 2**:

$$\begin{aligned} \textbf{Scaled RI} &= f_{\text{sigmoid}}(T;\theta) \\ &\quad + \beta_{\text{G}_{2/3}} \cdot \textbf{GDGT-2/GDGT-3} \\ &\quad + \beta_{\text{NO}_3} \cdot \log_{10}(\mathbf{NO_3}) \cdot \mathbb{I}_{(\mathbf{NO_3}<1.77)} + \epsilon \\ \epsilon &\sim \mathcal{N}(0, \sigma^2) \end{aligned} \tag{6}$$

$$\begin{aligned} \textbf{Scaled RI} = &\left[b + \frac{1 - b}{(1 + Q \cdot e^{-k(T - T_0)})^{1/\nu}} \right] \\ &+ \beta_{\text{G}_{2/3}} \cdot \textbf{GDGT-2/GDGT-3} \\ &+ \beta_{\text{NO}_3} \cdot \log_{10}(\mathbf{NO_3}) \cdot \mathbb{I}_{(\mathbf{NO_3}<1.77)} \end{aligned} \tag{7}$$

The temperature response parameters (θ_T: T_0, b, k, ν, Q) inherit informative priors from the top layer's posterior distributions:

$$\theta_T \sim \mathcal{N}(\text{median}(\theta_{T_{\text{culmeso}}}), \sigma^2_{\theta_{T_{\text{culmeso}}}}) \tag{8}$$

The model then solves for the posterior distribution of θ_T using gridded core-top data, allowing the environmental data to refine the temperature-RI relationship established by cultures and mesocosms. The sensitivities of scaled RI to AOA ecology ($\beta_{\text{G}_{2/3}}$) and nitrate concentrations (β_{NO_3}) are constrained exclusively by the core-top data, with the nutrient effect applying only below the low-nitrate threshold of 1.77 μmol/L. While the framework accommodates both SST and thermo-T formulations, we focus primarily on

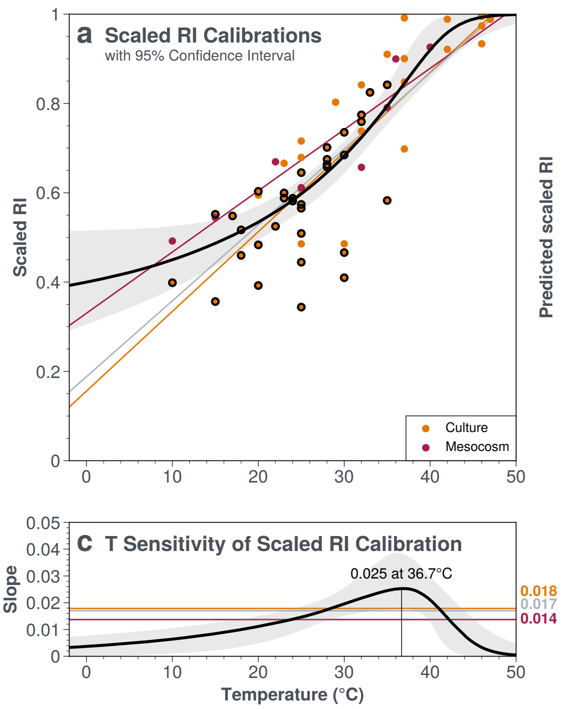

Fig. 6. Top layer of the TEXAS-Bay hierarchical model using culture and mesocosm data. (a) Scaled RI-temperature calibrations showing culture (orange dots) and mesocosm (pink dots) observations with the median sigmoid fit (black solid line) and 95% confidence interval (gray shading). Additional curves show separate linear regressions for culture (orange) and mesocosm (red) datasets. **(b)** Model performance metrics showing predicted versus measured scaled RI values (r^2 = 0.726, RMSE = 0.094). Posterior parameter estimates with 95% confidence intervals are displayed for the five-parameter generalized logistic function. Note: Culture data with black edges represent Marine Group 1.1a cultures. **(c)** Temperature sensitivity of the scaled RI calibration, expressed as the derivative (slope) of the median sigmoid curve. Maximum sensitivity (0.025) occurs at 36.7°C. Horizontal lines indicate constant slopes from linear regression fits: combined culture-mesocosm (gray, 0.017), mesocosm only (red, 0.014), and culture only (orange, 0.018).

thermo-T results in subsequent analyses. Prior and posterior distributions for all parameters of TEXAS-Bay are shown in **Appendix A** for both thermo-T (**Fig. A1**) and SST (**Fig. A2**).

7.2 Model Results

7.2.1 Performance Comparison: Bayesian versus Frequentist

The full multivariate thermo-T model explains 79.4% of the variance in core-top scaled RI data, with a median RMSE of 0.049 RI units (**Fig. 7b, 8h**). This represents a 17% improvement over the frequentist multivariate model ($r^2 = 0.777$; *Table S4*), demonstrating the benefit of the Bayesian hierarchical framework. The posterior distributions for the sigmoid parameters (θ_T: $T_0 = 31.465 \pm 4.091$, $b = 0.438 \pm 0.015$, $k = 0.0230 \pm 0.068$, $\nu = 3.033 \pm 1.040$, $Q = 2.669 \pm 1.843$) indicate that the inclusion of core-top data shifted the inflection point cooler by approximately 4°C relative to the culture-mesocosm priors ($T_{0_{\mathrm{culmeso}}} = 35.093$°C), while maintaining the fundamental sigmoidal geometry (**Fig. 7a–b**).

The secondary environmental controls exert distinct, quantifiable effects that corroborate the frequentist multivariate regression (**Section 6.2**) while revealing critical differences in parameter stability. The GDGT-2/GDGT-3 ratio, which tracks AOA ecological variations between shallow and deep-water assemblages (Rattanasriampaipong et al., 2022), exhibits a negative relationship with scaled RI ($\beta_{\mathrm{G}_{2/3}} = -0.006 \pm 0.001$). In the TEXAS-Bay forward model, this environmental correction shifts the calibration curve downward for high GDGT-2/GDGT-3 samples (**Fig. 7a**, dashed line shows correction for GDGT-2/GDGT-3 = 15), compensating for the cooling bias introduced by deep-water AOA contributions to the sedimentary signal. Critically, the Bayesian coefficient remains stable across both bivariate and multivariate formulations ($\beta_{\mathrm{G}_{2/3}} \approx -0.006$; *Table S4*), whereas the frequentist estimate fluctuates substantially with model specification ($\beta_{\mathrm{G}_{2/3}} = -0.0013$ to -0.0026). This stability highlights a key advantage of the Bayesian framework: simultaneous optimization of all parameters across multiple predictors, rather than anchoring the temperature term while sequentially fitting secondary variables.

Nitrate concentrations below 1.77 μmol/L exhibit a similar negative relationship ($\beta_{\mathrm{NO}_3} = -0.029 \pm 0.005$). This coefficient shifts the calibration curve upward in low nitrate regions (**Fig. 7a**, dash-dot line shows correction for $\mathrm{NO}_3 = 0.01$ μmol/L), consistent with the hypothesis that archaea increase cyclopentane ring numbers to maintain membrane fluidity under nutrient-limited conditions (Rattanasriampaipong et al., 2025). The relative magnitudes of these coefficients indicate that nitrate effects dominate in nutrient-depleted regions, while GDGT-2/GDGT-3 variations exert secondary but measurable influence where deep-water AOA contributions are significant.

Temperature sensitivity in the bottom layer deviates from the top layer, with maximum sensitivity (0.020) occurring at 30.7°C compared to 36.7°C in the culture-mesocosm dataset (**Fig. 7c**). This ∼6°C shift demonstrates how the Bayesian framework integrates sedimentary data to refine laboratory-derived parameters, capturing the complexity of natural archives where mixed assemblages and preservation effects modulate the temperature signal.

7.2.2 Sigmoid functional form resolves first-order spatial biases

Existing TEX_{86} calibrations exhibit persistent spatial biases across the global core-top dataset, revealing inadequacies in both functional form and the treatment of non-thermal environmental influences (**Fig. 8**). The original linear calibration (Schouten et al., 2002) and the logarithmic $\mathrm{TEX}_{86}^{\mathrm{H}}$ (Kim et al., 2010) both systematically underpredict TEX_{86} in polar/subpolar regions and the Red Sea, producing positive proxy residuals (observed − predicted TEX_{86}; **Fig. 8a**–**b**). When inverted for temperature reconstruction, these positive TEX_{86} residuals translate to warm biases in cold-water and nutrient-limited environments (**Section 8.1**, **Fig. 9a**–**b**). Conversely, subtropical and tropical sites—particularly in the Atlantic basin—exhibit negative TEX_{86} residuals where the models overpredict TEX_{86} (**Fig. 8a**–**b**), resulting in cold temperature biases (**Fig. 9a**–**b**). These consistent geographical biases indicate fundamental limitations in existing calibration models rather than random scatter, as confirmed by poor overall performance metrics ($r^2 = 0.616$ and 0.598, $RMSE = 0.079$ and 0.081, respectively; **Fig. 8a**–**b**).

The spatially-varying Bayesian calibration, BAYSPAR (Tierney & Tingley, 2014), significantly improves global regression metrics ($r^2 = 0.748$) by allowing for regionally flexible slope-intercept combinations. This localized approach pulls outliers, such as the Red Sea, closer to the 1:1 line (Fig. 8c) by adapting to local core-top characteristics. However, regional residual patterns persist, most notably a warm bias in polar regions (Fig. 8c). This is likely a result of both the exclusion of samples from >70°N in the BAYSPAR cal-

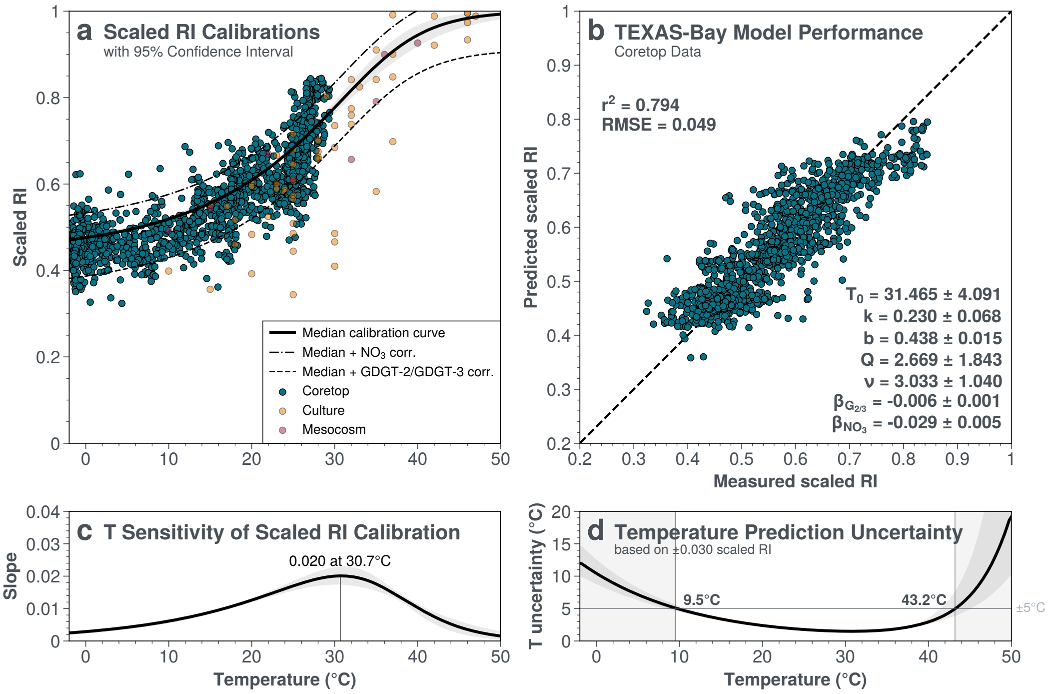

Fig. 7. Bottom layer of the TEXAS-Bay hierarchical model incorporating environmental covariates with core-top data. (a) Scaled RI-thermoT calibrations showing core-top sediment data (teal dots) alongside culture (orange dots) and mesocosm (pink dots) observations. The median sigmoid fit (solid black line) represents the temperature component with additional curves represent linear corrections of secondary predictors: GDGT-2/GDGT-3 = 15 (dashed line) and (NO_3) = 0.01 (dash-dot line). **(b)** A scatter plot of predicted versus measured scaled RI (r^2 = 0.794, RMSE = 0.049). Posterior distributions of model parameters with 95% confidence intervals are displayed. **(c)** Temperature sensitivity of the bottom layer calibration, expressed as the derivative of the median sigmoid curve. **(d)** Temperature reconstruction uncertainty as a function of temperature, calculated by propagating analytical uncertainty ($\pm$0.030 scaled RI) through the inverse calibration. Gray shading indicates regions where uncertainty exceeds $\pm 5°$C ($<$9.5°C and $>$43.2°C).

ibration set and the inherent limitations of using linear segments to capture a nonlinear temperature response at thermal extremes. Furthermore, BAYSPAR's subsurface variant (subT) shows reduced performance on the global dataset ($r^2 = 0.596$; Fig. 8d), highlighting the difficulty of reconciling TEX_{86} with any single depth metric through empirical regression alone.

The univariate TEXAS-Bay thermo-T model—incorporating the sigmoidal temperature function without nonthermal corrections—substantially reduces high-latitude biases, yielding homogeneous, low-magnitude residuals in polar regions (Fig. 8e). Notably, it achieves performance comparable to BAYSPAR-SST ($r^2 = 0.739$ vs. 0.748; *Table S4*) despite utilizing the complete global core-top dataset without the need for spatial subsetting or regional parameter tuning. This suggests that the sigmoidal functional form effectively captures the universal, nonlinear behavior of membrane lipid adaptation across the full thermal spectrum.

Unlike linear or logarithmic approximations, the S-curve represents the bounded physiological response of GDGT cyclization. At the lower thermal limit, marine archaea synthesize GDGTs with minimal cyclization. In the mid-range, cyclization accelerates, as observed in Marine Group I.1a cultures and seawater mesocosms (Fig. 6a). Finally, at extreme temperatures, cyclization approaches an upper asymptote—a saturation effect well-documented in terrestrial and hot-spring archaeal cultures (Fig. 6a). While marine AOA cultures have not yet demonstrated this saturation—as modern ocean temperatures remain far below the predicted upper limit—this universal temperature response is expected to govern the lipid distributions of the greenhouse oceans of the geologic past.

However, the univariate model still exhibits residual spatial structure, with positive biases particularly evident in oligotrophic regions such as the Red Sea (**Fig. 8e**). These patterns indicate that while the sigmoid resolves first-order functional form issues, nonthermal environmental factors still exert quantifiable influences on the global core-top dataset—motivating the environmental corrections described in the following section.

7.2.3 Nonthermal corrections improve model predictability

Evaluating TEXAS-Bay across progressive model configurations—from univariate (thermo-T only) to bivariate cases and the full multivariate model (thermo-T + GDGT-2/GDGT-3 + NO_3)—reveals systematic improvements in predictive performance for the global core-top dataset (**Fig. 8e–h**). The full multivariate model achieves $r^2 = 0.794$ and RMSE = 0.049, outperforming all existing TEX_{86} calibrations.

The two environmental covariates target distinct sources of residual variance (*Fig. S9*). The GDGT-2/GDGT-3 correction reduces scatter in the low-RI range (**Fig. 8f**), allowing the model to predict scaled RI values below the lower asymptote established by temperature alone ($b = 0.438$; **Fig. 7**). Deep-water archaea produce lipids with fewer cyclopentane rings (i.e., lower GDGT-2/GDGT-3 ratios) in cooler mesopelagic waters (Taylor et al., 2013); their export to surface sediments introduces a systematic cold bias. By accounting for the GDGT-2/GDGT-3 ratio, the model resolves this subsurface contribution, with the strongest effects in the tropical Atlantic and Pacific subtropical gyres (blue dots in *Fig. S9a*).

The nitrate correction targets the high-RI tail (**Fig. 8g**), accounting for values that exceed temperature-only predictions. Oligotrophic conditions impose nutrient stress on ammonia-oxidizing archaea, driving excess GDGT cyclization beyond the temperature-dependent baseline; this introduces a systematic warm bias in nutrient-depleted waters. By incorporating upper-ocean nitrate concentrations, the model eliminates systematic positive residuals that persist in univariate models and traditional TEX_{86} calibrations, with the strongest effects in the Red Sea and Mediterranean Sea, regions where nutrient effects on sedimentary TEX_{86} have been recently documented (red dots in *Fig. S9b*; Rattanasriampaipong et al., 2025).

The full multivariate model homogenizes residuals globally (**Fig. 8h**) through mechanistic rather than statistical improvements. TEXAS-Bay explicitly parameterizes biological and ecological processes causing systematic deviations from temperature-only relationships, rather than fitting curves to specific regions. This convergence is visible in predicted-versus-measured plots (**Fig. 8**, right column), where data at both thermal extremes move toward the 1:1 line. While forward modeling with SST yielded similar qualitative improvements (*Table S4*), thermo-T provided optimal fit, confirming that TEXAS-Bay successfully isolates temperature signal from environmental noise.

8 Applications to Paleothermometry

The primary utility of TEXAS-Bay lies in inverting the forward model to reconstruct temperatures from observed proxy data, following established Bayesian calibration frameworks for TEX_{86} and other paleotemperature proxies (Tierney & Tingley, 2014, 2018; Tierney et al., 2019). Beyond functional form limitations, existing calibrations like

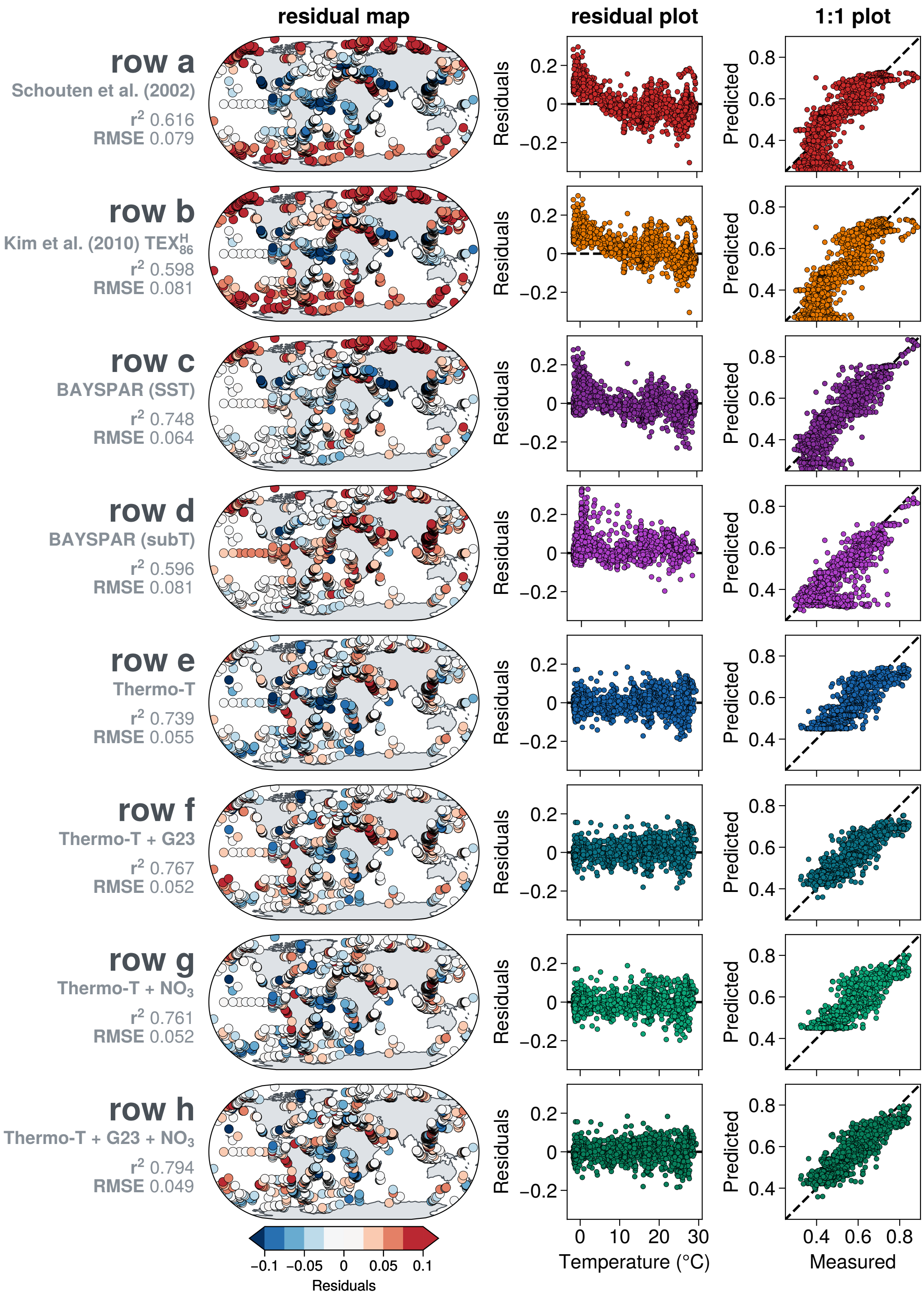

Fig. 8. Spatial distribution of residuals and performance metrics comparing traditional TEX_{86} calibrations, BAYSPAR, and TEXAS-Bay models. Left column displays global maps of residuals (predicted − measured); center column shows residuals plotted against temperature; right column shows observed versus predicted values with 1:1 dashed lines. **row (a)** Schouten et al. (2002) original linear calibration. **(b)** Kim et al. (2010) TEX_{86^H}. **(c)** BAYSPAR SST. **(d)** BAYSPAR subT. **(e)** TEXAS-Bay univariate Thermo-T model. **(f)** TEXAS-Bay bivariate model with GDGT-2/GDGT-3. **(g)** TEXAS-Bay bivariate model with nitrate. **(h)** TEXAS-Bay full multivariate model. Note: G23 denotes GDGT-2/GDGT-3.

$\mathrm{TEX}_{86}^{\mathrm{H}}$ rely on inverse regression, where temperature is regressed onto the proxy index ($T = f(\mathrm{Index})$) (Kim et al., 2008, 2010; Z. Liu et al., 2009; O'Brien et al., 2017) (see *Table S2*). This approach suffers from *regression dilution bias*: measurement error in the predictor variable (the Index) systematically flattens the regression slope, distorting sensitivity estimates when extrapolating beyond the modern calibration range.

8.1 Inversion Performance for Modern Core-Top Temperatures

By forward modeling the proxy as a function of environmental variables ($\mathrm{RI} = f(T)$) and mathematically inverting this relationship, TEXAS-Bay preserves the true physiological sensitivity and properly propagates uncertainties. Since our model is multivariate, inverting to predict ocean temperatures (thermo-T or SST) requires constraints on AOA ecology and nitrate levels. GDGT-2/GDGT-3 ratios are directly available from GDGT measurements, while quantitative paleo-nitrate proxies have yet to be established.

We evaluated TEXAS-Bay inversion performance by subsampling calibration parameters from the forward model posterior distributions and generating ensemble temperature predictions for the global core-top dataset. We compared these predictions against traditional calibration models using temperature residual maps, residual plots, and 1:1 comparisons (**Fig. 9**).

All models show RMSE values of 4.76–6.16°C, with BAYSPAR-SST achieving the lowest RMSE (4.76°C) for the extended core-top dataset. All models exhibit warm temperature residuals (observed − predicted) in the Arctic, while Southern Ocean warm biases only disappear in BAYSPAR models. The TEXAS-Bay multivariate model resolves Red Sea warm biases (similar to BAYSPAR-SST) by incorporating nitrate as a secondary predictor, whereas other models show persistent warm biases in this region. While TEXAS-Bay substantially improves forward prediction of proxy values (∼38% RMSE reduction), this improvement translates more modestly to inverse temperature reconstructions (**Fig. 9**). Median predictions from TEXAS-Bay consistently overpredict thermo-T, producing warm biases in polar regions visually similar to the original linear (Schouten et al., 2002) and logarithmic $\mathrm{TEX}_{86}^{\mathrm{H}}$ models (Kim et al., 2010).

However, the warm biases at high latitudes in TEXAS-Bay reconstructions likely arise from nonlinear error propagation inherent to the sigmoid functional form. Propagating typical analytical uncertainty (±0.030 scaled RI) through the inverse calibration produces temperature uncertainties exceeding ±5°C at temperatures <9.5°C and >43.2°C (**Fig. 7d**), where the flattened sigmoid response exhibits extremely low temperature sensitivity (**Fig. 7c**). In these asymptotic regions, small changes in scaled RI amplify into large temperature uncertainties. This mechanistic limitation reflects genuine proxy system constraints rather than calibration failure. The reduced precision at temperature extremes (**Fig. 9h**) thus represents an honest characterization of proxy resolving power, suggesting that published GDGT-based temperature reconstructions <9.5°C and >43.2°C potentially carry larger uncertainties than previously recognized.

8.2 TEXAS-Bay Applications to Quaternary and Paleogene Climate Records

To evaluate TEXAS-Bay performance in paleoclimate applications, we applied the framework to two key intervals representing contrasting climate boundary conditions: the late Quaternary glacial-interglacial cycles (0–150 ka; **Section 8.2.1**) and the Paleocene-Eocene Thermal Maximum (PETM; ∼56 Ma; **Section 8.2.2**). These intervals span the

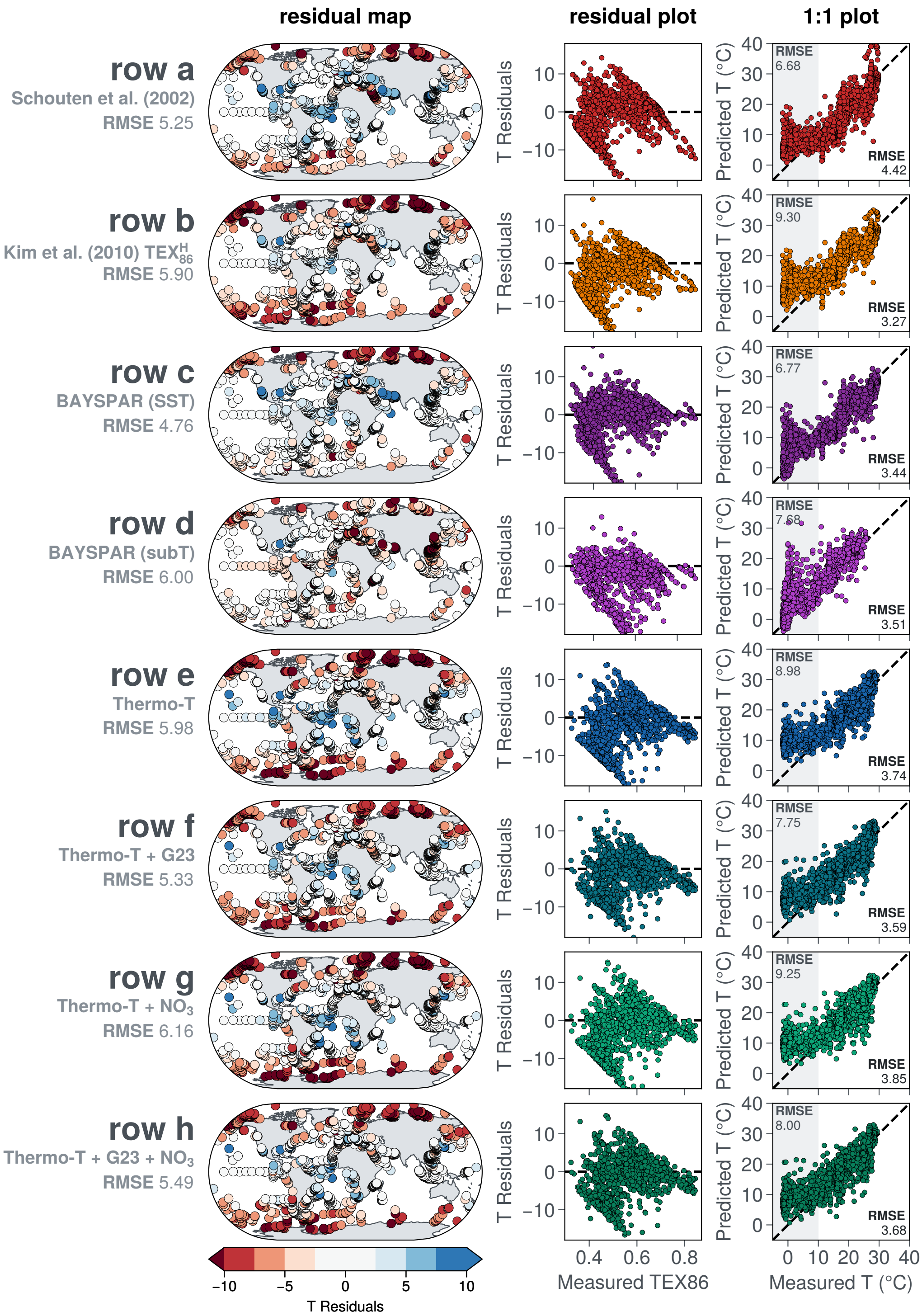

Fig. 9. Spatial distribution of residuals and performance metrics for inverse temperature predictions from traditional TEX_{86} calibrations, BAYSPAR, and TEXAS-Bay models. Left column: global maps of residuals (predicted − measured temperature); center column: temperature residuals versus measured TEX_{86}; right column: predicted versus measured temperature with 1:1 dashed lines. **(a)** Schouten et al. (2002) original linear calibration. **(b)** Kim et al. (2010) TEX_{86^H}. **(c)** BAYSPAR SST. **(d)** BAYSPAR subT. **(e)** TEXAS-Bay univariate Thermo-T model. **(f)** TEXAS-Bay bivariate model with GDGT-2/GDGT-3. **(g)** TEXAS-Bay bivariate model with nitrate. **(h)** TEXAS-Bay full multivariate model. All models show inverse predictions (temperature from TEX_{86} or Ring Index). G23 denotes GDGT-2/GDGT-3.

range from near-modern conditions to extreme greenhouse warmth, providing critical tests of the TEXAS-Bay across different oceanographic regimes and temperature ranges.

For both case studies, we calculated scaled Ring Index values from published GDGT distributions and applied Mahalanobis distance (D_M) screening (**Section 5.1**) to identify samples with anomalous GDGT assemblages. Samples exceeding the D_M screening threshold were flagged as potentially influenced by non-thermal factors. We applied the TEXAS-Bay bivariate model (thermo-T + GDGT-2/GDGT-3) rather than the full multivariate model, as GDGT-2/GDGT-3 ratios are directly available from GDGT measurements while paleo-nitrate concentrations remain unconstrained for most paleoclimate records.

8.2.1 Glacial–Interglacial Cycles: Evaluating Depth Habitat Constraints

We applied the TEXAS-Bay temperature reconstruction framework to five published marine sediment records spanning the last 150 kyr from the Western Pacific Warm Pool (WPWP) and mid- to high-latitude settings in the Tasman Sea region (**Fig. 10**). A fundamental challenge in reconstructing ocean temperatures for these sites is whether sedimentary TEX_{86} signals record sea surface temperature (SST) or subsurface temperature (SubT). Previous studies have relied on regional calibration datasets, modern oceanographic context, or comparisons against independent paleoclimate proxies to guide the choice of calibration model (Schouten et al., 2013; Ho et al., 2014; Elling et al., 2025). This decision becomes particularly critical in tropical versus high-latitude settings, where AOA depth habitats and thermal stratification differ markedly.

Published temperature reconstructions from WPWP sites (MD97-2151, MD98-2152) and the equatorial Indian Ocean (U1482) predominantly employ subsurface calibrations (Panels 1–3 in **Fig. 10**; Yamamoto et al., 2013; Windler et al., 2019; ?, ?). Studies predating BAYSPAR typically used depth-specific calibrations based on regional core-top datasets. For example, at Site MD97-2151 in the South China Sea, Yamamoto et al. (2013) reconstructed subT using TEX^{H}_{86} calibrated to ocean temperatures at the 30–125 m depth interval, following evidence that TEX^{H}_{86} correlates more strongly with subsurface temperatures in this region (Jia et al., 2012). This approach addresses concerns about geographic variability in TEX_{86}–temperature relationships and reflects hypotheses regarding AOA habitat depth in tropical oligotrophic waters. Modern oceanographic studies suggest that in stratified tropical oceans, marine AOA populations often concentrate at the base of the euphotic zone near the deep chlorophyll maximum (DCM), where they integrate thermal signals from the upper water column rather than exclusively from the sea surface (Huguet et al., 2007). More recent work applies BAYSPAR SST or subT variants depending on regional oceanography and interpretation of AOA habitat depth (Tierney & Tingley, 2014).

TEXAS-Bay reconstructions using the thermo-T + GDGT-2/GDGT-3 bivariate model closely track these published subT records (Panels 1–3 in **Fig. 10**), showing particularly strong agreement with BAYSPAR subT reconstructions. This concordance suggests that the TEXAS-Bay bivariate model effectively captures the depth-integrated thermal habitat of marine AOA in stratified tropical waters. Notably, both subT and thermo-T reconstructions at these sites show clear separation from SST estimates, mirroring the temperature stratification observed in modern ocean profiles.

In contrast, mid- to high-latitude sites in the Tasman Sea (DSDP591, U1510) demonstrate convergence between subT, thermo-T, and SST estimates, reflecting weaker thermal stratification in these well-mixed water columns (Panels 4–5 in **Fig. 10**). Published TEX_{86}-based reconstructions at these locations yield temperatures substantially warmer

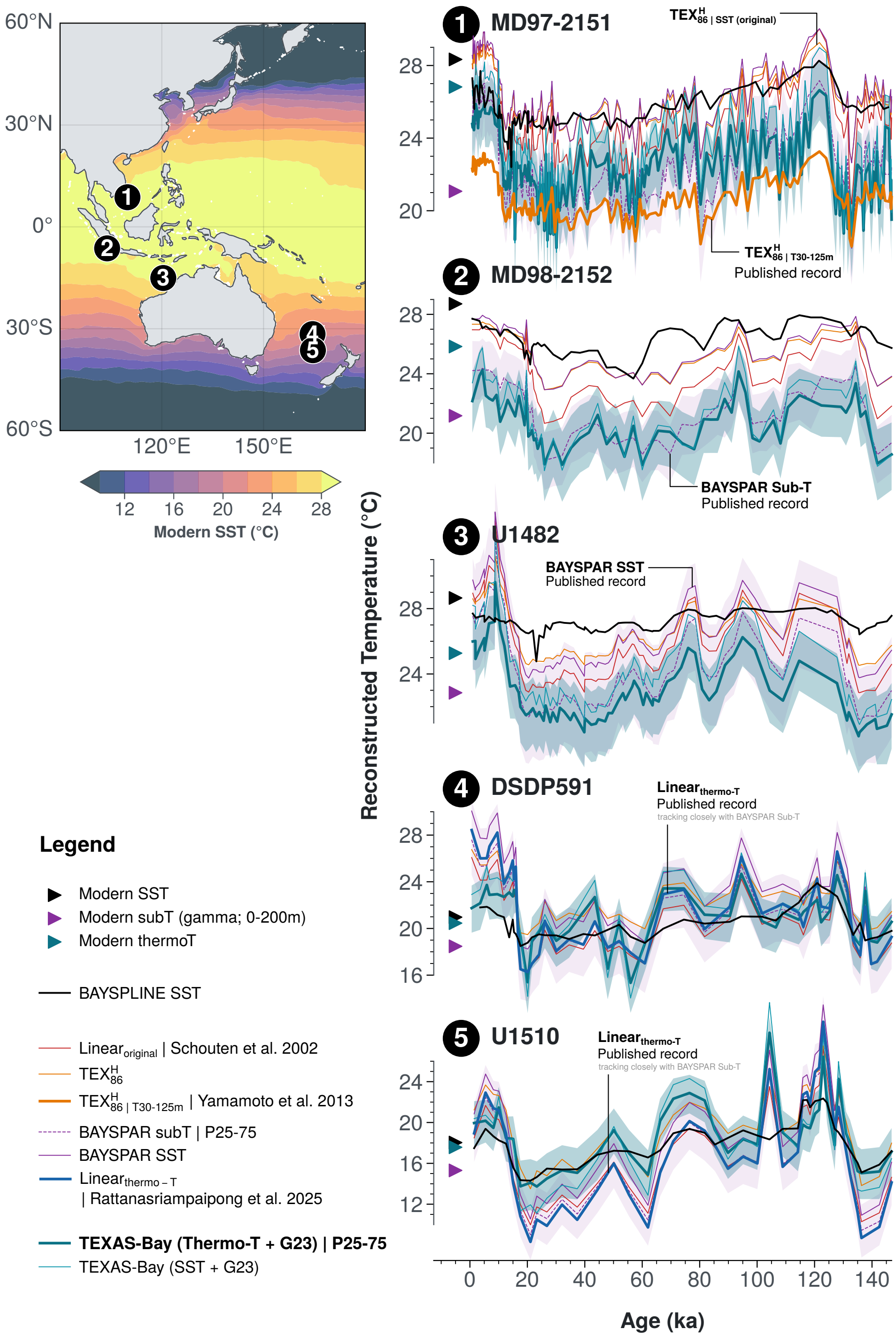

Fig. 10. Comparison of TEXAS-Bay temperature reconstructions with published records across the Western Pacific and Eastern Indian Ocean. Five marine sediment cores spanning tropical to mid-latitude settings show reconstructed temperatures over the last 150 kyr. TEXAS-Bay with thermo-T + GDGT-2/GDGT-3 (teal), TEXAS-Bay with SST + GDGT-2/GDGT-3 (blue), BAYSPAR SST (pink solid) and SubT (pink dashed), TEX_{86}^{H} (orange), and $U_{37}^{k'}$ BAYSPLINE SST (black) are compared with published reconstructed T records. Triangles show modern temperatures: SST (black), gamma-weighted subsurface 0–200 m (purple), and thermo-T (teal). Sites include: (1) MD97-2151 (Yamamoto et al., 2013); (2) MD98-2152 (Windler et al., 2019); (3) U1482 (?, ?); (4) DSDP591 (Rattanasriampaipong et al., 2025); (5) U1510 (Rattanasriampaipong et al., 2025).

than modern observations and $U^{k'}_{37}$ BAYSPLINE SST estimates, with discrepancies exceeding 4–6°C during interglacial periods (Rattanasriampaipong et al., 2025). TEXAS-Bay thermo-T + GDGT-2/GDGT-3 reconstructions reduce this warm bias during the Holocene, yielding temperatures more consistent with modern thermo-T values and BAYSPLINE SST reconstructions. During glacial periods, TEXAS-Bay and $U^{k'}_{37}$ reconstructions converge at both sites, with only sporadic divergences, suggesting that SST and thermo-T generally track each other when thermal stratification is minimal (as observed in modern conditions where SST and thermo-T show small differences).

However, short-lived warm excursions exceeding 10°C persist in TEXAS-Bay reconstructions throughout both records (Panels 4–5 in **Fig. 10**). These anomalies, previously identified by Rattanasriampaipong et al. (2025) as potential artifacts of nutrient stress affecting AOA lipid distributions, remain present despite incorporating GDGT-2/GDGT-3 as an ecological covariate. This observation suggests that while GDGT-2/GDGT-3 ratios capture broad ecological gradients related to AOA community composition, additional information on paleo-nutrient levels may be necessary to fully resolve temperature biases arising from physiological responses to nutrient stress.

8.2.2 Paleocene-Eocene Thermal Maximum: Reconciling Proxy-Model Discrepancies

The Paleocene-Eocene Thermal Maximum (PETM; ∼56 Ma) represents the most severe and abrupt global warming event of the Cenozoic, characterized by a ∼5‰ carbon isotope excursion and widespread ocean warming (McInerney & Wing, 2011). Reconstructing absolute PETM temperatures is critical for constraining Earth system sensitivity to carbon cycle perturbations and validating climate model simulations of extreme greenhouse conditions (cf. Tierney et al., 2022). TEX_{86} has been widely applied to reconstruct PETM SST (Zachos et al., 2006; Sluijs et al., 2007; Frieling et al., 2017), yet different calibrations yield temperature estimates spanning >10°C at individual sites (Hollis et al., 2019), hampering proxy-model comparisons and climate sensitivity assessments.

We applied TEXAS-Bay to published GDGT data from two well-studied PETM sections: Wilson Lake (New Jersey coastal plain; 39°N paleolatitude; Zachos et al., 2006; Sluijs et al., 2007) and ODP Site 959 (equatorial Atlantic; 3°N paleolatitude; Frieling et al., 2017). Calibration choice dramatically affects reconstructed temperatures at both sites (**Fig. 11**, panels 1–2). At Wilson Lake, the published SST record (Zachos et al., 2006; Sluijs et al., 2007) uses a modified linear calibration fit to core-top data with SST >20°C (*Table S2*; Schouten et al., 2003), yielding pre-PETM baseline temperatures of 24–26°C and peak PETM SST of ∼32–34°C. The original linear model and BAYSPAR-SST produce substantially warmer temperatures: ∼28–30°C for the pre-PETM baseline and ∼40–44°C during peak PETM. TEX^{H}_{86} yields pre-PETM SST similar to linear calibrations while producing intermediate peak PETM temperatures. At ODP959, linear calibrations (Schouten et al., 2002; Tierney & Tingley, 2014) yield pre-PETM baselines of 35–37°C and peak PETM temperatures exceeding 43–46°C, while TEX^{H}_{86} (Kim et al., 2010) yields cooler estimates (pre-PETM: ∼33–34°C; peak PETM: ∼37–38°C).

TEXAS-Bay SST reconstructions closely track TEX^{H}_{86} at both sites, yielding pre-PETM baselines of ∼24–26°C at Wilson Lake and ∼32–34°C at ODP959, with peak PETM temperatures of ∼32–36°C and ∼37–38°C, respectively. These pre-PETM baselines align with climate model predictions for 3× preindustrial CO_2 (Zhu et al., 2019)—i.e. the widely accepted PETM background state (Hollis et al., 2019; Tierney et al., 2022)—while peak PETM temperatures fall within the 6×–9× CO_2 range (Zhu et al., 2019). Consequently,

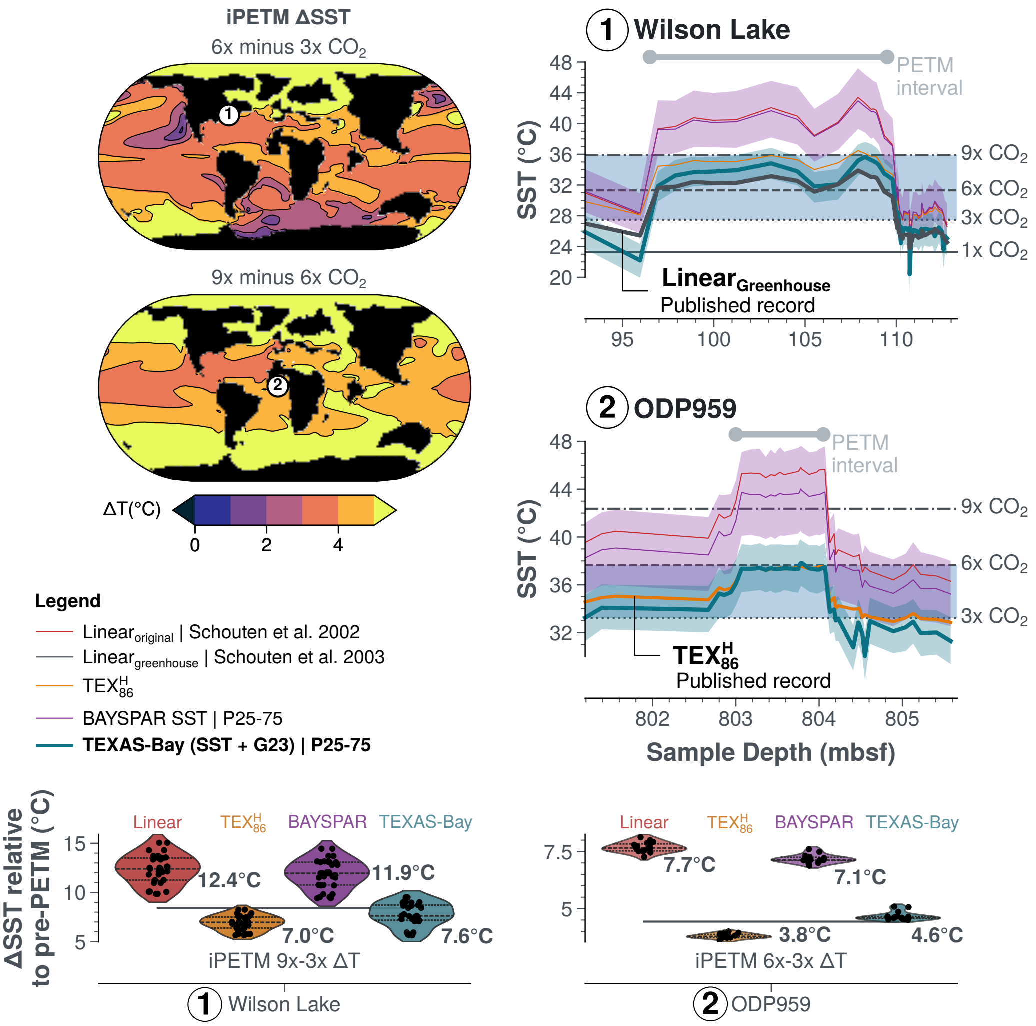

Fig. 11. Comparison of TEXAS-Bay with published TEX_{86} calibrations for PETM temperature reconstructions and climate model predictions. Left panels: Sea surface temperature anomalies (ΔSST) from isotope-enabled PETM climate model experiments (Zhu et al., 2019): 6× minus 3× CO_2 (top) and 9× minus 6× CO_2 (bottom), with ODP959 (equatorial Atlantic) and Wilson Lake (mid-latitude North Atlantic) marked. **Right panels:** Proxy SST reconstructions across the PETM at (1) ODP959 (Frieling et al., 2017) and (2) Wilson Lake (Sluijs et al., 2007; Zachos et al., 2006) using $\text{Linear}_{\text{original}}$ (Schouten et al., 2002), $\text{Linear}_{\text{greenhouse}}$ (Schouten et al., 2003), TEX_{86}^{H} (Kim et al., 2010), BAYSPAR SST (Tierney & Tingley, 2014), and TEXAS-Bay (SST + GDGT-2/GDGT-3). Horizontal dashed lines indicate climate model SST for 1×, 3×, 6×, and 9× preindustrial CO_2. **Bottom panels:** Violin plots of ΔSST relative to pre-PETM baseline, with median values annotated. TEXAS-Bay yields PETM warming estimates (7.6°C at Wilson Lake; 4.6°C at ODP959) intermediate between TEX_{86}^{H} and linear calibrations, consistent with climate model predictions for high-CO_2 greenhouse conditions.

TEXAS-Bay reduces estimated PETM warming magnitude (ΔSST relative to pre-PETM) by 4–7°C compared to linear calibrations, with median warming of 7.6°C at Wilson Lake (9×–3× CO_2 scenario) and 4.6°C at ODP959 (6×–3× CO_2 scenario) (**Fig. 11**, bottom panels). Linear calibrations produce warming magnitudes of 12.4°C (Wilson Lake) and 7.7°C (ODP959) that exceed even 9× CO_2 simulations.

The improved proxy-model agreement from TEXAS-Bay and TEX_{86}^{H} arises from nonlinear functional forms that exhibit reduced temperature sensitivity at high temperatures. Linear calibrations extrapolating beyond the modern core-top range produce artificial temperature inflation, whereas sigmoid (TEXAS-Bay) and logarithmic (TEX_{86}^{H}) forms naturally dampen this effect (**Fig. 7c** and **Fig. 1**). However, these approaches differ fundamentally: TEXAS-Bay derives reduced sensitivity from forward modeling with proper Bayesian inversion, whereas TEX_{86}^{H} relies on inverse regression subject to dilution bias, with its asymptotic behavior mathematically compressing large TEX_{86} changes into small temperature variations at high values (>0.7).

9 Conclusions

We developed TEXAS, a proxy system model framework that integrates temperature, AOA ecology, and nutrient availability to predict marine archaeal GDGT distributions. TEXAS-Bay, the corresponding temperature calibration, predicts ocean temperatures from observed GDGT data by modeling the scaled Ring Index as a sigmoid function of temperature with linear corrections for AOA community composition (GDGT-2/GDGT-3) and nutrient availability. By constraining the functional form using culture and mesocosm experiments and applying Bayesian calibration to an extended global core-top dataset, TEXAS-Bay achieves substantial improvements in forward prediction compared to existing TEX_{86} calibrations. Applications to paleoclimate records spanning Quaternary glacial-interglacial cycles and the Paleocene-Eocene Thermal Maximum demonstrate that TEXAS-Bay reconstructions closely track published records across diverse climate states when calibrated using all regions of the global core-top dataset, including high-latitude sites typically excluded from existing calibrations.

TEXAS-Bay advances TEX_{86} paleothermometry in three key areas. *First*, incorporating nonthermal covariates reduces forward prediction errors, enabling temperature reconstructions across diverse oceanographic settings without requiring *a priori* selection between SST and subsurface calibrations. *Second*, the constrained sigmoid functional form prevents artificial temperature inflation when extrapolating to greenhouse warmth while providing realistic uncertainty estimates that reflect reduced proxy sensitivity at temperature extremes (<9.5°C and >43.2°C). *Third*, Mahalanobis distance screening provides objective criteria for identifying anomalous GDGT distributions. However, temperature reconstructions beyond the modern calibration range remain inherently uncertain due to reduced proxy sensitivity and significantly fewer constraints from laboratory experiments. Moreover, persistent warm excursions in Tasman Sea paleorecords suggest that additional paleo-nutrient proxies are needed to fully resolve nutrient stress effects.

Future refinement will benefit from expanded core-top compilations from undersampled regions, experimental studies quantifying AOA physiological responses to environmental variables, and integration with emerging proxies for paleo-productivity and redox conditions. By explicitly accounting for ecological and physiological controls on GDGT biosynthesis within a Bayesian framework, TEXAS provides a unified, mechanistically grounded approach applicable from modern oceans to ancient greenhouse climates. This framework enables more robust proxy-model comparisons essential for constraining Earth system sensitivity and validating climate projections under high-CO_2 scenarios.

Appendix A Bayesian Regression Model Priors

Priors for the Bayesian regression parameters

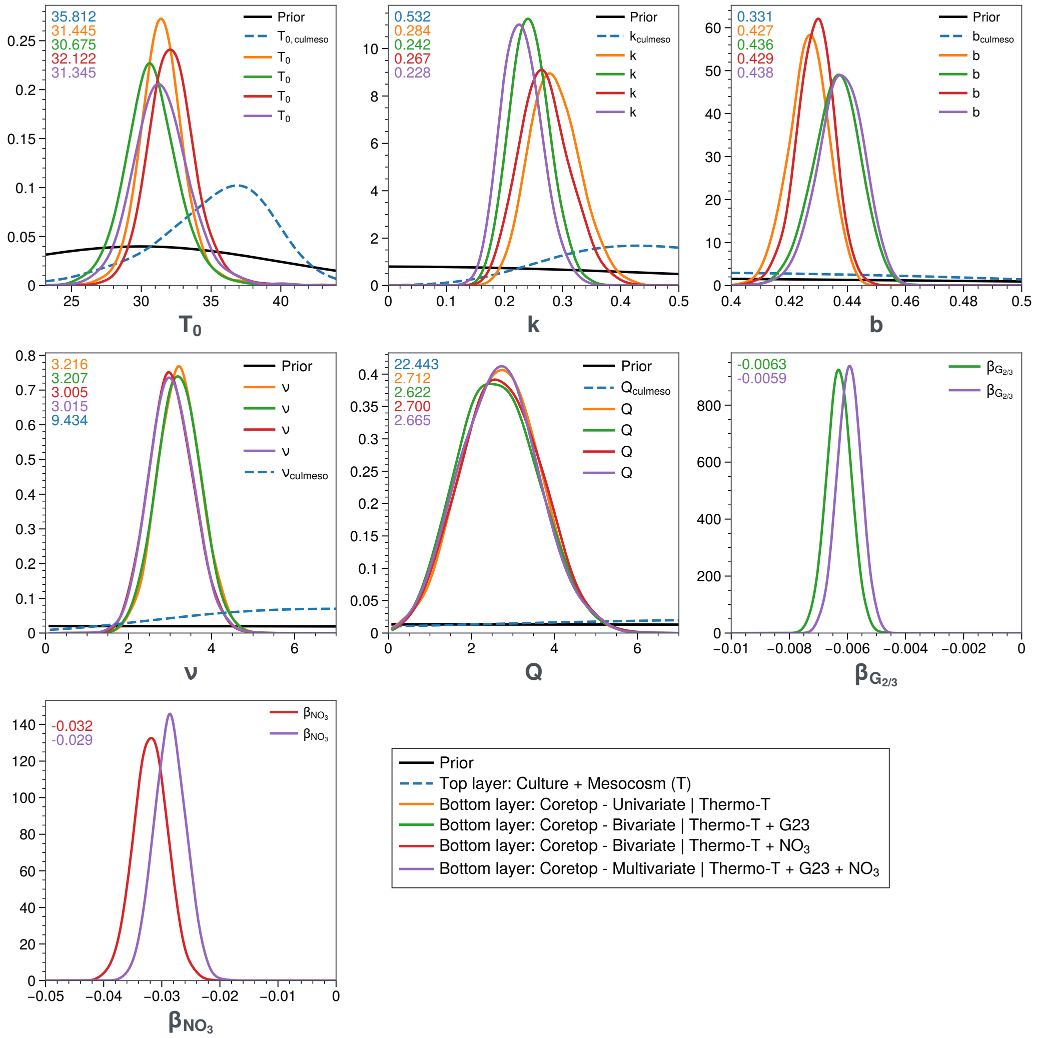

Fig. A1. Prior and posterior distributions for TEXAS-Bay model parameters for thermo-T. Distributions show the evolution from weakly informative priors (black) to posteriors estimated from hierarchical Bayesian models with varying covariate structures. Bottom layer posteriors (colored solid lines) represent coretop-only calibrations under four specifications: univariate temperature-only (orange), bivariate with GDGT-2/GDGT-3 (green), bivariate with nitrate (red), and multivariate with both covariates (purple). Priors of the bottom layer are informed by top layer posteriors (blue; dashed) represent the combined culture and mesocosm dataset with temperature only. Parameters include: T_0 (lower asymptote temperature, $^\circ$C), k (steepness parameter), b (inflection point), ν (asymmetry parameter), Q (upper asymptote parameter), $\beta_{\mathrm{G2/3}}$ (GDGT-2/GDGT-3 regression coefficient), and β_{NO3} (nitrate regression coefficient). Substantial deviations of posteriors from priors confirm that data dominate inference. The near-identical posteriors for sigmoid parameters (T_0, k, b, ν) across culture/mesocosm and coretop datasets demonstrate convergence in Ring Index space, supporting the unified calibration framework.

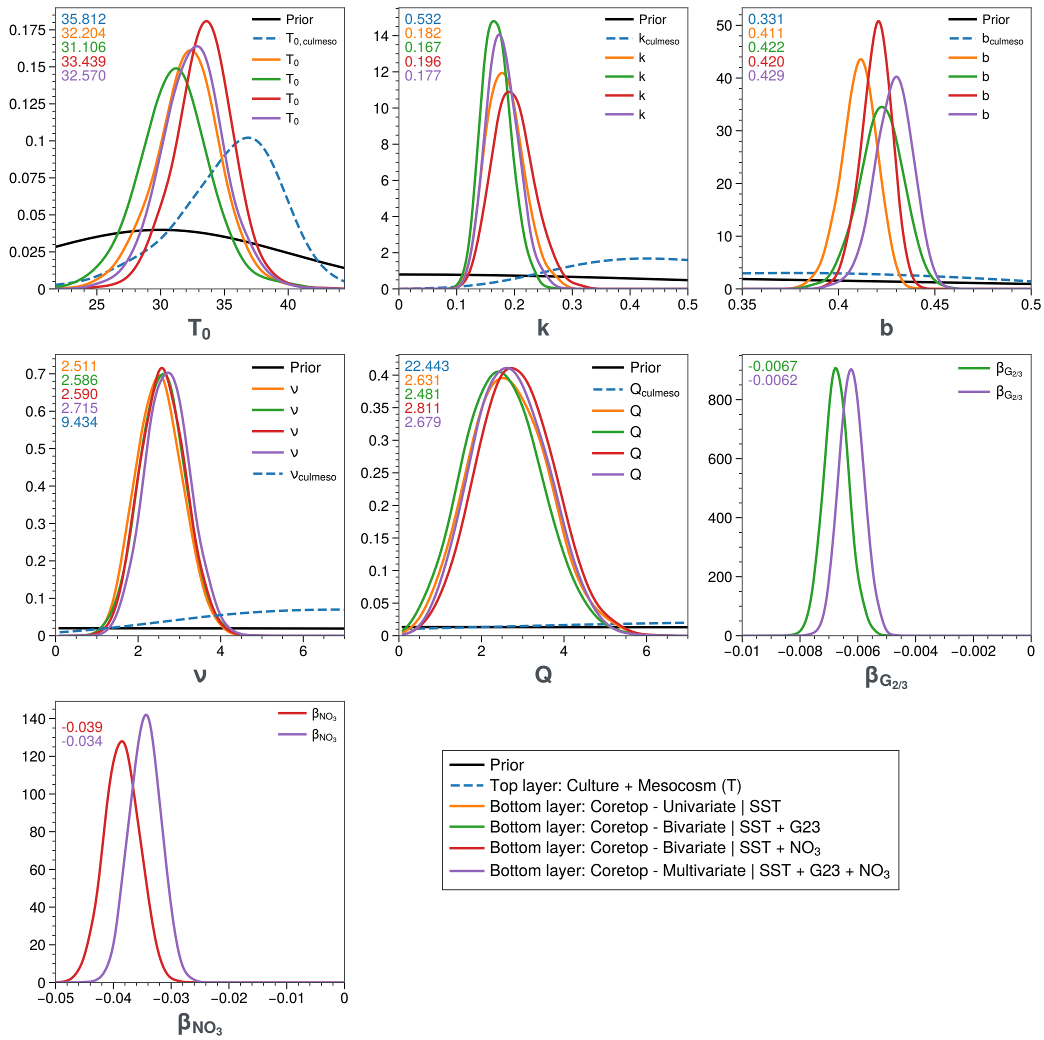

Fig. A2. Prior and posterior distributions for TEXAS-Bay model parameters for SST.

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors.

Acknowledgments
This work was supported by the NOAA Climate and Global Change Postdoctoral Fellowship Program, administered by UCAR's Cooperative Programs for the Advancement of Earth System Science (CPAESS) under the NOAA Science Collaboration Program award (Grant number #NA23OAR4310383B) to R.R. J.E.T. acknowledges support from the David and Lucile Packard Fellowship in Science and Engineering and the Thomas R. Brown Distinguished Chair in Integrative Science.

References

Bale, N. J., Palatinszky, M., Rijpstra, W. I. C., Herbold, C. W., Wagner, M., Damsté, J. S. S., & Atomi, H. (2019). Membrane lipid composition of the moderately thermophilic ammonia-oxidizing archaeon "candidatus nitrosotenuis uzonensis" at different growth temperatures. *Applied and Environmental Microbiology*, *85*, e01332-19. doi: doi:10.1128/AEM.01332-19

Bale, N. J., Villanueva, L., Hopmans, E. C., Schouten, S., & Sinninghe Damsté, J. S. (2013). Different seasonality of pelagic and benthic thaumarchaeota in the north sea. *Biogeosciences*, *10*, 7195-7206. Retrieved from https://bg.copernicus.org/articles/10/7195/2013/ doi: 10.5194/bg-10-7195-2013

Blaga, C. I., Reichart, G.-J., Heiri, O., & Sinninghe Damsté, J. S. (2009). Tetraether membrane lipid distributions in water-column particulate matter and sediments: a study of 47 european lakes along a north–south transect. *Journal of Paleolimnology*, *41*, 523-540.

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., ... Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*, 1-32. Retrieved from https://www.jstatsoft.org/index.php/jss/article/view/v076i01 doi: 10.18637/jss.v076.i01

Castañeda, I. S., Schefuß, E., Pätzold, J., Sinninghe Damsté, J. S., Weldeab, S., & Schouten, S. (2010). Millennial-scale sea surface temperature changes in the eastern mediterranean (nile river delta region) over the last 27,000 years. *Paleoceanography*, *25*. Retrieved from https://doi.org/10.1029/2009PA001740 doi: https://doi.org/10.1029/2009PA001740

Ceccopieri, M., Carreira, R. S., Wagener, A. L. R., Hefter, J. H., & Mollenhauer, G. (2018). On the application of alkenone- and gdgt-based temperature proxies in the south-eastern brazilian continental margin. *Organic Geochemistry*, *126*, 43-56. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638018302407 doi: https://doi.org/10.1016/j.orggeochem.2018.10.009

Chazen, C. R. (2011). *Holocene climate evolution of the eastern tropical pacific told from high resolution climate records from the peru margin and equa-*

torial upwelling regions (Doctoral dissertation, Brown University). doi: https://doi.org/10.7301/Z0D50K7P

Chen, J., Hu, P., Li, X., Yang, Y., Song, J., Li, X., … Lü, X. (2018). Impact of water depth on the distribution of igdgts in the surface sediments from the northern south china sea: applicability of tex86 in marginal seas. *Frontiers of Earth Science*, *12*, 95-107. Retrieved from `https://doi.org/10.1007/s11707-016-0620-1` doi: 10.1007/s11707-016-0620-1

Chen, W., Mohtadi, M., Schefuß, E., & Mollenhauer, G. (2014). Organic-geochemical proxies of sea surface temperature in surface sediments of the tropical eastern indian ocean. *Deep Sea Research Part I: Oceanographic Research Papers*, *88*, 17-29. Retrieved from `http://www.sciencedirect.com/science/article/pii/S0967063714000417` doi: https://doi.org/10.1016/j.dsr.2014.03.005

Cumberland, W. N., Fong, Y., Yu, X., Defawe, O., Frahm, N., & Rosa, S. D. (2015). Nonlinear calibration model choice between the four and five-parameter logistic models. *Journal of Biopharmaceutical Statistics*, *25*, 972-983. (PMID: 24918306) doi: 10.1080/10543406.2014.920345

De Rosa, M., Esposito, E., Gambacorta, A., Nicolaus, B., & Bu'Lock, J. D. (1980). Effects of temperature on ether lipid composition of ¡em¿Caldariella acidophila¡/em¿. *Phytochemistry*, *19*(5), 827–831. Retrieved from `http://www.sciencedirect.com/science/article/pii/003194228085120X` doi: 10.1016/0031-9422(80)85120-X

Dunkley Jones, T., Eley, Y. L., Thomson, W., Greene, S. E., Mandel, I., Edgar, K., & Bendle, J. A. (2020). Optimal: a new machine learning approach for gdgt-based palaeothermometry. *Clim. Past*, *16*, 2599-2617. Retrieved from `https://cp.copernicus.org/articles/16/2599/2020/https://cp.copernicus.org/articles/16/2599/2020/cp-16-2599-2020.pdf` doi: 10.5194/cp-16-2599-2020

Elling, F. J., Inglis, G. N., Davtian, N., Hurley, S., Naafs, B. D., O'Connor, L. K., … Zhao, X. (2025, nov). Archaeal tetraether lipids as tracers for past marine environmental change. *ESS Open Archive*. Retrieved from `http://dx.doi.org/10.22541/essoar.176409484.44403346/v1` doi: 10.22541/essoar.176409484.44403346/v1

Elling, F. J., Könneke, M., Lipp, J. S., Becker, K. W., Gagen, E. J., & Hinrichs, K.-U. (2014). Effects of growth phase on the membrane lipid composition of the thaumarchaeon Nitrosopumilus maritimus and their implications for archaeal lipid distributions in the marine environment. *Geochimica et Cosmochimica Acta*, *141*, 579–597. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0016703714004670` doi: https://doi.org/10.1016/j.gca.2014.07.005

Elling, F. J., Könneke, M., Mußmann, M., Greve, A., & Hinrichs, K.-U. (2015). Influence of temperature, pH, and salinity on membrane lipid composition and TEX86 of marine planktonic thaumarchaeal isolates. *Geochimica et Cosmochimica Acta*, *171*, 238–255. Retrieved from `http://www.sciencedirect.com/science/article/pii/S0016703715005335` doi: https://doi.org/10.1016/j.gca.2015.09.004

Elling, F. J., Könneke, M., Nicol, G. W., Stieglmeier, M., Bayer, B., Spieck, E., … Hinrichs, K.-U. (2017). Chemotaxonomic characterisation of the thaumarchaeal lipidome. *Environmental Microbiology*, *19*(7), 2681–2700. Retrieved from `https://sfamjournals.onlinelibrary.wiley.com/doi/abs/10.1111/`

1462-2920.13759 doi: https://doi.org/10.1111/1462-2920.13759

Evans, M. N., Tolwinski-Ward, S. E., Thompson, D. M., & Anchukaitis, K. J. (2013). Applications of proxy system modeling in high resolution paleoclimatology. *Quaternary Science Reviews*, *76*, 16-28. Retrieved from https://www.sciencedirect.com/science/article/pii/S0277379113002011 doi: https://doi.org/10.1016/j.quascirev.2013.05.024

Evans, T. W., Könneke, M., Lipp, J. S., Adhikari, R. R., Taubner, H., Elvert, M., & Hinrichs, K.-U. (2018). Lipid biosynthesis of nitrosopumilus maritimus dissected by lipid specific radioisotope probing (lipid-rip) under contrasting ammonium supply. *Geochimica et Cosmochimica Acta*, *242*, 51-63. Retrieved from https://www.sciencedirect.com/science/article/pii/S0016703718305064 doi: https://doi.org/10.1016/j.gca.2018.09.001

Fallet, U., Castañeda, I. S., Henry-Edwards, A., Richter, T. O., Boer, W., Schouten, S., & Brummer, G.-J. (2012). Sedimentation and burial of organic and inorganic temperature proxies in the mozambique channel, sw indian ocean. *Deep Sea Research Part I: Oceanographic Research Papers*, *59*, 37-53. Retrieved from https://www.sciencedirect.com/science/article/pii/S0967063711001907https://linkinghub.elsevier.com/retrieve/pii/S0967063711001907 doi: 10.1016/j.dsr.2011.10.002

Frieling, J., Gebhardt, H., Huber, M., Adekeye, O. A., Akande, S. O., Reichart, G.-J., ... Sluijs, A. (2017). Extreme warmth and heat-stressed plankton in the tropics during the paleocene-eocene thermal maximum. *Science Advances*, *3*, e1600891. Retrieved from https://doi.org/10.1126/sciadv.1600891 doi: 10.1126/sciadv.1600891

Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (1995). *Bayesian data analysis.* Chapman and Hall/CRC.

Gottschalk, P. G., & Dunn, J. R. (2005). The five-parameter logistic: A characterization and comparison with the four-parameter logistic. *Analytical Biochemistry*, *343*, 54-65. doi: https://doi.org/10.1016/j.ab.2005.04.035

Hagemann, J. R., Lembke-Jene, L., Lamy, F., Vorrath, M.-E., Kaiser, J., Müller, J., ... Tiedemann, R. (2023). *Collection of gdgt-derived temperatures of 22 new sites at the chilean margin, supplemented by various previously published sites [data set].* PANGAEA. Retrieved from https://doi.org/10.1594/PANGAEA.964270 doi: 10.1594/PANGAEA.964270

Harning, D. J., Andrews, J. T., Belt, S. T., Cabedo-Sanz, P., Geirsdottir, A., Dildar, N., ... Sepúlveda, J. (2019). Sea ice control on winter subsurface temperatures of the north iceland shelf during the little ice age: A tex86 calibration case study. *Paleoceanography and Paleoclimatology*, *34*, 1006-1021. Retrieved from https://doi.org/10.1029/2018PA003523 doi: 10.1029/2018PA003523

Harning, D. J., Holman, B., Woelders, L., Jennings, A. E., & Sepúlveda, J. (2023). Biomarker characterization of the north water polynya, baffin bay: implications for local sea ice and temperature proxies. *Biogeosciences*, *20*, 229-249. Retrieved from https://doi.org/10.1594/PANGAEA.956212https://bg.copernicus.org/articles/20/229/2023/ doi: 10.5194/bg-20-229-2023

Hernández-Sánchez, M. T., Woodward, E. M. S., Taylor, K. W. R., Henderson, G. M., & Pancost, R. D. (2014). Variations in gdgt distributions through the water column in the south east atlantic ocean. *Geochimica et Cosmochimica Acta*, *132*, 337-348. Retrieved from http://www.sciencedirect.com/science/article/pii/S0016703714000970 doi: https://doi.org/10.1016/j.gca.2014.02.009

Ho, S. L., Mollenhauer, G., Fietz, S., Martínez-Garcia, A., Lamy, F., Rueda, G., ... Tiedemann, R. (2014). Appraisal of tex86 and tex86l thermometries in subpolar and polar regions. *Geochimica et Cosmochimica Acta*, *131*, 213-226. Retrieved from https://www.sciencedirect.com/science/article/pii/S0016703714000337 doi: https://doi.org/10.1016/j.gca.2014.01.001

Ho, S. L., Yamamoto, M., Mollenhauer, G., & Minagawa, M. (2011). Core top tex86 values in the south and equatorial pacific. *Organic Geochemistry*, *42*, 94-99. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638010002639 doi: https://doi.org/10.1016/j.orggeochem.2010.10.012

Hollis, C. J., Jones, T. D., Anagnostou, E., Bijl, P. K., Cramwinckel, M. J., Cui, Y., ... Lunt, D. J. (2019, 7). The deepmip contribution to pmip4: methodologies for selection, compilation and analysis of latest paleocene and early eocene climate proxy data, incorporating version 0.1 of the deepmip database. *Geoscientific Model Development*, *12*, 3149-3206. Retrieved from https://gmd.copernicus.org/articles/12/3149/2019/ doi: 10.5194/gmd-12-3149-2019

Hopmans, E. C., Weijers, J. W. H., Schefuß, E., Herfort, L., Sinninghe Damsté, J. S., & Schouten, S. (2004). A novel proxy for terrestrial organic matter in sediments based on branched and isoprenoid tetraether lipids. *Earth and Planetary Science Letters*, *224*, 107-116. Retrieved from http://www.sciencedirect.com/science/article/pii/S0012821X04003176 doi: https://doi.org/10.1016/j.epsl.2004.05.012

Hu, J., Meyers, P. A., Chen, G., Peng, P., & Yang, Q. (2012). Archaeal and bacterial glycerol dialkyl glycerol tetraethers in sediments from the eastern lau spreading center, south pacific ocean. *Organic Geochemistry*, *43*, 162-167. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638011002956 doi: https://doi.org/10.1016/j.orggeochem.2011.10.012

Huguet, C., Schimmelmann, A., Thunell, R., Lourens, L. J., Damsté, J. S. S., & Schouten, S. (2007). A study of the tex86 paleothermometer in the water column and sediments of the santa barbara basin, california. *Paleoceanography*, *22*. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2006PA001310 doi: 10.1029/2006pa001310

Hurley, S. J., Elling, F. J., Könneke, M., Buchwald, C., Wankel, S. D., Santoro, A. E., ... Pearson, A. (2016). Influence of ammonia oxidation rate on thaumarchaeal lipid composition and the TEX86 temperature proxy. *Proceedings of the National Academy of Sciences of the U.S.A.*, *113*(28), 7762–7767. Retrieved from https://www.pnas.org/content/pnas/113/28/7762.full.pdf doi: 10.1073/pnas.1518534113

Inglis, G. N., Farnsworth, A., Lunt, D., Foster, G. L., Hollis, C. J., Pagani, M., ... Pancost, R. D. (2015). Descent toward the icehouse: Eocene sea surface cooling inferred from gdgt distributions. *Paleoceanography*, *30*, 1000-1020. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2014PA002723 doi: 10.1002/2014pa002723

Inglis, G. N., & Tierney, J. E. (2020). *The tex86 paleotemperature proxy.* Cambridge University Press. Retrieved from https://www.cambridge.org/core/elements/tex86-paleotemperature-proxy/71294DD2B56EFD6BA6B7D92E7E95D63C doi: DOI:10.1017/9781108846998

Jaeschke, A., Wengler, M., Hefter, J., Ronge, T. A., Geibert, W., Mollenhauer, G., ... Lamy, F. (2017). A biomarker perspective on dust, productivity, and sea surface temperature in the pacific sector of the southern ocean.

Geochimica et Cosmochimica Acta, *204*, 120-139. Retrieved from https://www.sciencedirect.com/science/article/pii/S0016703717300595 doi: https://doi.org/10.1016/j.gca.2017.01.045

Jia, G., Zhang, J., Chen, J., Peng, P., & Zhang, C. L. (2012). Archaeal tetraether lipids record subsurface water temperature in the south china sea. *Organic Geochemistry*, *50*, 68-77. Retrieved from http://www.sciencedirect.com/science/article/pii/S0146638012001465 doi: https://doi.org/10.1016/j.orggeochem.2012.07.002

Judd, E. J., Tierney, J. E., Huber, B. T., Wing, S. L., Lunt, D. J., Ford, H. L., … Zhang, Y. G. (2022). The PhanSST global database of Phanerozoic sea surface temperature proxy data. *Scientific Data*, *9*(1). doi: 10.1038/s41597-022-01826-0

Judd, E. J., Tierney, J. E., Lunt, D. J., Montañez, I. P., Huber, B. T., Wing, S. L., & Valdes, P. J. (2024). A 485-million-year history of Earth's surface temperature. *Science*, *385*(6715), eadk3705. Retrieved from https://doi.org/10.1126/science.adk3705 doi: 10.1126/science.adk3705

Jung, M.-Y., Kim, J.-G., Damsté, J. S. S., Rijpstra, W. I. C., Madsen, E. L., Kim, S.-J., … Rhee, S.-K. (2016). A hydrophobic ammonia-oxidizing archaeon of the nitrosocosmicus clade isolated from coal tar-contaminated sediment. *Environmental Microbiology Reports*, *8*, 983-992. Retrieved from https://enviromicro-journals.onlinelibrary.wiley.com/doi/abs/10.1111/1758-2229.12477 doi: https://doi.org/10.1111/1758-2229.12477

Jung, M.-Y., Park, S.-J., Kim, S.-J., Kim, J.-G., Damsté, J. S. S., Jeon, C. O., & Rhee, S.-K. (2014). A mesophilic, autotrophic, ammonia-oxidizing archaeon of thaumarchaeal group i.1a cultivated from a deep oligotrophic soil horizon. *Applied and Environmental Microbiology*, *80*, 3645-3655. Retrieved from https://journals.asm.org/doi/abs/10.1128/aem.03730-13 doi: 10.1128/AEM.03730-13

Jung, M.-Y., Park, S.-J., Min, D., Kim, J.-S., Rijpstra, W. I. C., Damsté, J. S. S., … Rhee, S.-K. (2011). Enrichment and characterization of an autotrophic ammonia-oxidizing archaeon of mesophilic crenarchaeal group i.1a from an agricultural soil. *Applied and Environmental Microbiology*, *77*, 8635-8647. Retrieved from https://journals.asm.org/doi/abs/10.1128/aem.05787-11 doi: 10.1128/AEM.05787-11

Kaiser, J., Ruggieri, N., Hefter, J., Siegel, H., Mollenhauer, G., Arz, H. W., & Lamy, F. (2014). Lipid biomarkers in surface sediments from the gulf of genoa, ligurian sea (nw mediterranean sea) and their potential for the reconstruction of palaeo-environments. *Deep Sea Research Part I: Oceanographic Research Papers*, *89*, 68-83. Retrieved from https://www.sciencedirect.com/science/article/pii/S0967063714000661 doi: https://doi.org/10.1016/j.dsr.2014.04.009

Kaiser, J., Schouten, S., Kilian, R., Arz, H. W., Lamy, F., & Sinninghe Damsté, J. S. (2015). Isoprenoid and branched gdgt-based proxies for surface sediments from marine, fjord and lake environments in chile. *Organic Geochemistry*, *89-90*, 117-127. Retrieved from https://www.sciencedirect.com/science/article/pii/S014663801500193X doi: 10.1016/j.orggeochem.2015.10.007

Karner, M. B., DeLong, E. F., & Karl, D. M. (2001). Archaeal dominance in the mesopelagic zone of the pacific ocean. *Nature*, *409*, 507-510. Retrieved from http://dx.doi.org/10.1038/35054051

Kim, J.-H., Schouten, S., Hopmans, E. C., Donner, B., & Sinninghe Damsté, J. S.

(2008). Global sediment core-top calibration of the tex86 paleothermometer in the ocean. *Geochimica et Cosmochimica Acta*, *72*, 1154-1173. Retrieved from http://www.sciencedirect.com/science/article/pii/S0016703707007132 doi: https://doi.org/10.1016/j.gca.2007.12.010

Kim, J.-H., van der Meer, J., Schouten, S., Helmke, P., Willmott, V., Sangiorgi, F., ... Sinninghe Damsté, J. S. (2010). New indices and calibrations derived from the distribution of crenarchaeal isoprenoid tetraether lipids: Implications for past sea surface temperature reconstructions. *Geochimica et Cosmochimica Acta*, *74*(16), 4639–4654. Retrieved from http://www.sciencedirect.com/science/article/pii/S0016703710003054 doi: https://doi.org/10.1016/j.gca.2010.05.027

Kim, J.-H., Villanueva, L., Zell, C., & Sinninghe Damsté, J. S. (2016). Biological source and provenance of deep-water derived isoprenoid tetraether lipids along the portuguese continental margin. *Geochimica et Cosmochimica Acta*, *172*, 177-204. Retrieved from http://www.sciencedirect.com/science/article/pii/S0016703715005396 doi: https://doi.org/10.1016/j.gca.2015.09.010

Kusch, S., Rethemeyer, J., Hopmans, E. C., Wacker, L., & Mollenhauer, G. (2016). Factors influencing 14c concentrations of algal and archaeal lipids and their associated sea surface temperature proxies in the black sea. *Geochimica et Cosmochimica Acta*, *188*, 35-57. Retrieved from https://www.sciencedirect.com/science/article/pii/S001670371630268X doi: https://doi.org/10.1016/j.gca.2016.05.025

Lamping, N., Müller, J., Hefter, J., Mollenhauer, G., Haas, C., Shi, X., ... Hillenbrand, C.-D. (2021). Evaluation of lipid biomarkers as proxies for sea ice and ocean temperatures along the antarctic continental margin. *Climate of the Past*, *17*, 2305-2326. Retrieved from https://cp.copernicus.org/articles/17/2305/2021/ doi: 10.5194/cp-17-2305-2021

Leider, A., Hinrichs, K.-U., Mollenhauer, G., & Versteegh, G. J. M. (2010). Core-top calibration of the lipid-based u37k′ and tex86 temperature proxies on the southern italian shelf (sw adriatic sea, gulf of taranto). *Earth and Planetary Science Letters*, *300*, 112-124. Retrieved from https://www.sciencedirect.com/science/article/pii/S0012821X10006291 doi: https://doi.org/10.1016/j.epsl.2010.09.042

Lengger, S. K., Hopmans, E. C., Sinninghe Damsté, J. S., & Schouten, S. (2014). Fossilization and degradation of archaeal intact polar tetraether lipids in deeply buried marine sediments (peru margin). *Geobiology*, *12*, 212-220. Retrieved from https://onlinelibrary.wiley.com/doi/abs/10.1111/gbi.12081 doi: https://doi.org/10.1111/gbi.12081

Liu, X.-L., Zhu, C., Wakeham, S. G., & Hinrichs, K.-U. (2014). In situ production of branched glycerol dialkyl glycerol tetraethers in anoxic marine water columns. *Marine Chemistry*, *166*, 1-8. Retrieved from https://www.sciencedirect.com/science/article/pii/S0304420314001388 doi: https://doi.org/10.1016/j.marchem.2014.08.008

Liu, Z., Pagani, M., Zinniker, D., DeConto, R., Huber, M., Brinkhuis, H., ... Pearson, A. (2009). Global cooling during the eocene-oligocene climate transition. *Science*, *323*, 1187-1190. Retrieved from https://science.sciencemag.org/content/sci/323/5918/1187.full.pdf doi: 10.1126/science.1166368

Lo, L., Belt, S. T., Lattaud, J., Friedrich, T., Zeeden, C., Schouten, S., ... Hodell, D. A. (2018). Precession and atmospheric co2 modulated variability of sea ice in the central okhotsk sea since 130,000 years ago. *Earth and Planetary Sci-*

ence Letters, *488*, 36-45. Retrieved from https://www.sciencedirect.com/science/article/pii/S0012821X1830061X doi: https://doi.org/10.1016/j.epsl.2018.02.005

Locarnini, R. A., Mishonov, A. V., Baranova, O. K., Reagan, J. R., Boyer, T. P., Seidov, D., ... Dukhovskoy, D. (2024). *World ocean atlas 2023, volume 1: Temperature* (Tech. Rep.). Retrieved from https://repository.library.noaa.gov/view/noaa/60599 doi: https://doi.org/10.25923/54bh-1613

Lü, X., Yang, H., Song, J., Versteegh, G. J. M., Li, X., Yuan, H., ... Xie, S. (2014). Sources and distribution of isoprenoid glycerol dialkyl glycerol tetraethers (gdgts) in sediments from the east coastal sea of china: Application of gdgt-based paleothermometry to a shallow marginal sea. *Organic Geochemistry*, *75*, 24-35. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638014001570 doi: https://doi.org/10.1016/j.orggeochem.2014.06.007

McInerney, F. A., & Wing, S. L. (2011). The paleocene-eocene thermal maximum: A perturbation of carbon cycle, climate, and biosphere with implications for the future. *Annual Review of Earth and Planetary Sciences*, *39*(1), 489–516.

Nieto-Moreno, V., Martínez-Ruiz, F., Willmott, V., García-Orellana, J., Masqué, P., & Sinninghe Damsté, J. S. (2013). Climate conditions in the westernmost mediterranean over the last two millennia: An integrated biomarker approach. *Organic Geochemistry*, *55*, 1-10. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638012002252 doi: https://doi.org/10.1016/j.orggeochem.2012.11.001

O'Brien, C. L., Robinson, S. A., Pancost, R. D., Sinninghe Damsté, J. S., Schouten, S., Lunt, D. J., ... Wrobel, N. E. (2017). Cretaceous sea-surface temperature evolution: Constraints from TEX86 and planktonic foraminiferal oxygen isotopes. *Earth-Science Reviews*, *172*, 224–247. Retrieved from http://www.sciencedirect.com/science/article/pii/S0012825217303859 doi: https://doi.org/10.1016/j.earscirev.2017.07.012

Pan, A., Yang, Q., Zhou, H., Ji, F., Wang, H., & Pancost, R. D. (2016). A diagnostic gdgt signature for the impact of hydrothermal activity on surface deposits at the southwest indian ridge. *Organic Geochemistry*, *99*, 90-101.

Park, Y.-H., Yamamoto, M., Nam, S.-I., Irino, T., Polyak, L., Harada, N., ... Saitoh, S.-I. (2014). Distribution, source and transportation of glycerol dialkyl glycerol tetraethers in surface sediments from the western arctic ocean and the northern bering sea. *Marine Chemistry*, *165*, 10-24. Retrieved from https://www.sciencedirect.com/science/article/pii/S030442031400111X doi: https://doi.org/10.1016/j.marchem.2014.07.001

Pearson, A., Pi, Y., Zhao, W., Li, W., Li, Y., Inskeep, W., ... Zhang, C. L. (2008, jun). Factors Controlling the Distribution of Archaeal Tetraethers in Terrestrial Hot Springs. *Applied and Environmental Microbiology*, *74*(11), 3523–3532. Retrieved from https://aem.asm.org/content/aem/74/11/3523.full.pdfhttps://journals.asm.org/doi/10.1128/AEM.02450-07 doi: 10.1128/AEM.02450-07

Perruche, C. (2018). Product user manual for the global ocean biogeochemistry hindcast global_reanalysis_bio_001_029. version 1. *Copernicus Marine Environment Monitoring Service*. doi: http://dx.doi.org/10.25607/OBP-490

Pitcher, A., Hopmans, E. C., Mosier, A. C., Park, S.-J., Rhee, S.-K., Francis, C. A., ... Damsté, J. S. S. (2011). Core and intact polar glycerol dibiphytanyl glycerol tetraether lipids of ammonia-oxidizing archaea enriched from marine and estuarine sediments. *Applied and Environmental Microbiology*, *77*, 3468-3477. Retrieved from https://journals.asm.org/doi/abs/10.1128/aem.02758-10 doi: 10.1128/AEM.02758-10

Pitcher, A., Rychlik, N., Hopmans, E. C., Spieck, E., Rijpstra, W. I. C., Ossebaar, J., ... Damsté, J. S. S. (2010). Crenarchaeol dominates the membrane lipids of candidatus nitrososphaera gargensis, a thermophilic group i.1b archaeon. *The ISME Journal*, *4*, 542-552. Retrieved from https://doi.org/10.1038/ismej.2009.138 doi: 10.1038/ismej.2009.138

Qin, W., Carlson, L. T., Armbrust, E. V., Devol, A. H., Moffett, J. W., Stahl, D. A., & Ingalls, A. E. (2015). Confounding effects of oxygen and temperature on the TEX86 signature of marine Thaumarchaeota. *Proceedings of the National Academy of Sciences of the U.S.A.*, *112*(35), 10979–10984.

Ramanathan, V., & Collins, W. (1991). Thermodynamic regulation of ocean warming by cirrus clouds deduced from observations of the 1987 el niño. *Nature*, *351*, 27-32. Retrieved from https://doi.org/10.1038/351027a0 doi: 10.1038/351027a0

Rattanasriampaipong, R., Tierney, J. E., Abell, J. T., & Gilmore, L. D. (2025). A Nutrient Effect on the TEX86 Paleotemperature Proxy. *Geophysical Research Letters*, *52*(12), e2025GL115237. Retrieved from https://doi.org/10.1029/2025GL115237 doi: 10.1029/2025GL115237

Rattanasriampaipong, R., Zhang, Y. G., Pearson, A., Hedlund, B. P., & Zhang, S. (2022). Archaeal lipids trace ecology and evolution of marine ammonia-oxidizing archaea. *Proceedings of the National Academy of Sciences*, *119*(31), e2123193119. Retrieved from https://doi.org/10.1073/pnas.2123193119 doi: 10.1073/pnas.2123193119

Richards, F. J. (1959). A flexible growth function for empirical use. *Journal of Experimental Botany*, *10*, 290-300. Retrieved from http://www.jstor.org/stable/23686557

Richey, J. N., Hollander, D. J., Flower, B. P., & Eglinton, T. I. (2011). Merging late holocene molecular organic and foraminiferal-based geochemical records of sea surface temperature in the gulf of mexico. *Paleoceanography*, *26*. Retrieved from https://doi.org/10.1029/2010PA002000 doi: https://doi.org/10.1029/2010PA002000

Richey, J. N., & Tierney, J. E. (2016). Gdgt and alkenone flux in the northern gulf of mexico: Implications for the tex86 and uk'37 paleothermometers. *Paleoceanography*, *31*, 1547-1561. Retrieved from https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2016PA003032 doi: 10.1002/2016pa003032

Schouten, S., Forster, A., Panoto, F. E., & Sinninghe Damsté, J. S. (2007). Towards calibration of the TEX86 palaeothermometer for tropical sea surface temperatures in ancient greenhouse worlds. *Organic Geochemistry*, *38*(9), 1537–1546. Retrieved from http://www.sciencedirect.com/science/article/pii/S0146638007001209 doi: https://doi.org/10.1016/j.orggeochem.2007.05.014

Schouten, S., Hopmans, E. C., & Damsté, J. S. S. (2013). The organic geochemistry of glycerol dialkyl glycerol tetraether lipids: A review. *Organic Geochemistry*, *54*, 19-61. Retrieved from http://www.sciencedirect.com/science/article/pii/S0146638012001982 doi: https://doi.org/10.1016/j.orggeochem.2012.09.006

Schouten, S., Hopmans, E. C., Forster, A., van Breugel, Y., Kuypers, M. M. M., & Sinninghe Damsté, J. S. (2003). Extremely high sea-surface temperatures at low latitudes during the middle cretaceous as revealed by archaeal membrane lipids. *Geology*, *31*, 1069-1072. Retrieved from `https://doi.org/10.1130/G19876.1` doi: 10.1130/g19876.1

Schouten, S., Hopmans, E. C., Schefuss, E., & Sinninghe Damsté, J. S. (2002). Distributional variations in marine crenarchaeotal membrane lipids: a new tool for reconstructing ancient sea water temperatures? *Earth and Planetary Science Letters*, *204*(1-2), 265–274.

Schouten, S., Hopmans, E. C., van der Meer, J., Mets, A., Bard, E., Bianchi, T. S., … Damsté, J. S. S. (2009). An interlaboratory study of tex86 and bit analysis using high-performance liquid chromatography–mass spectrometry. *Geochemistry, Geophysics, Geosystems*, *10*. doi: https://doi.org/10.1029/2008GC002221

Seki, O., Bendle, J. A., Harada, N., Kobayashi, M., Sawada, K., Moossen, H., … Sakamoto, T. (2014). Assessment and calibration of tex86 paleothermometry in the sea of okhotsk and sub-polar north pacific region: Implications for paleoceanography. *Progress in Oceanography*, *126*, 254-266. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0079661114000615` doi: https://doi.org/10.1016/j.pocean.2014.04.013

Seki, O., Sakamoto, T., Sakai, S., Schouten, S., Hopmans, E. C., Sinninghe Damsté, J. S., & Pancost, R. D. (2009). Large changes in seasonal sea ice distribution and productivity in the sea of okhotsk during the deglaciations. *Geochemistry, Geophysics, Geosystems*, *10*. Retrieved from `https://doi.org/10.1029/2009GC002613` doi: https://doi.org/10.1029/2009GC002613

Shevenell, A. E., Ingalls, A. E., Domack, E. W., & Kelly, C. (2011). Holocene southern ocean surface temperature variability west of the antarctic peninsula. *Nature*, *470*, 250-254. Retrieved from `https://doi.org/10.1038/nature09751` doi: 10.1038/nature09751

Sinninghe Damsté, J. S. (2016). Spatial heterogeneity of sources of branched tetraethers in shelf systems: The geochemistry of tetraethers in the berau river delta (kalimantan, indonesia). *Geochimica et Cosmochimica Acta*, *186*, 13-31. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0016703716302022` doi: https://doi.org/10.1016/j.gca.2016.04.033

Sinninghe Damsté, J. S., Ossebaar, J., Schouten, S., & Verschuren, D. (2012). Distribution of tetraether lipids in the 25-ka sedimentary record of lake challa: extracting reliable tex86 and mbt/cbt palaeotemperatures from an equatorial african lake. *Quaternary Science Reviews*, *50*, 43-54. doi: https://doi.org/10.1016/j.quascirev.2012.07.001

Sinninghe Damsté, J. S., Schouten, S., Hopmans, E. C., van Duin, A. C. T., & Geenevasen, J. A. J. (2002). Crenarchaeol: the characteristic core glycerol dibiphytanyl glycerol tetraether membrane lipid of cosmopolitan pelagic crenarchaeota. *Journal of Lipid Research*, *43*, 1641-1651. doi: 10.1194/jlr.m200148-jlr200

Sinninghe Damsté, J. S., Warden, L. A., Berg, C., Jürgens, K., & Moros, M. (2022). Evaluation of the distributions of hydroxylated glycerol dibiphytanyl glycerol tetraethers (gdgts) in holocene baltic sea sediments for reconstruction of sea surface temperature: the effect of changing salinity. *Climate of the Past*, *18*, 2271-2288. Retrieved from `https://cp.copernicus.org/articles/18/2271/2022/` doi: 10.5194/cp-18-2271-2022

Sluijs, A., Brinkhuis, H., Schouten, S., Bohaty, S. M., John, C. M., Zachos, J. C., … Dickens, G. R. (2007). Environmental precursors to rapid light carbon injection at the palaeocene/eocene boundary. *Nature*, *450*, 1218-1221. Retrieved from https://doi.org/10.1038/nature06400 doi: 10.1038/nature06400

Smith, M., Deckker, P. D., Rogers, J., Brocks, J., Hope, J., Schmidt, S., ... Schouten, S. (2013). Comparison of u37k′, tex86h and ldi temperature proxies for reconstruction of south-east australian ocean temperatures. *Organic Geochemistry*, *64*, 94-104. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638013001915 doi: https://doi.org/10.1016/j.orggeochem.2013.08.015

Taylor, K. W. R., Huber, M., Hollis, C. J., Hernandez-Sanchez, M. T., & Pancost, R. D. (2013). Re-evaluating modern and Palaeogene GDGT distributions: Implications for SST reconstructions. *Global and Planetary Change*, *108*, 158–174. Retrieved from http://www.sciencedirect.com/science/article/pii/S0921818113001537 doi: https://doi.org/10.1016/j.gloplacha.2013.06.011

Tierney, J. E., Malevich, S. B., Gray, W., Vetter, L., & Thirumalai, K. (2019, 12). Bayesian calibration of the mg/ca paleothermometer in planktic foraminifera. *Paleoceanography and Paleoclimatology*, *34*, 2005-2030. Retrieved from https://doi.org/10.1029/2019PA003744 doi: https://doi.org/10.1029/2019PA003744

Tierney, J. E., & Tingley, M. P. (2014). A bayesian, spatially-varying calibration model for the tex86 proxy. *Geochimica et Cosmochimica Acta*, *127*, 83-106. Retrieved from http://www.sciencedirect.com/science/article/pii/S0016703713006650 doi: https://doi.org/10.1016/j.gca.2013.11.026

Tierney, J. E., & Tingley, M. P. (2015). A tex 86 surface sediment database and extended bayesian calibration. *Scientific Data*, *2*, 1-10.

Tierney, J. E., & Tingley, M. P. (2018, 3). Bayspline: A new calibration for the alkenone paleothermometer. *Paleoceanography and Paleoclimatology*, *33*, 281-301. Retrieved from https://doi.org/10.1002/2017PA003201 (https://doi.org/10.1002/2017PA003201) doi: https://doi.org/10.1002/2017PA003201

Tierney, J. E., Ummenhofer, C. C., & DeMenocal, P. B. (2015). Past and future rainfall in the horn of africa. *Science Advances*, *1*, e1500682. Retrieved from https://www.science.org/doi/abs/10.1126/sciadv.1500682 doi: 10.1126/sciadv.1500682

Tierney, J. E., Zhu, J., Li, M., Ridgwell, A., Hakim, G. J., Poulsen, C. J., ... Kump, L. R. (2022, 10). Spatial patterns of climate change across the paleocene–eocene thermal maximum. *Proceedings of the National Academy of Sciences*, *119*, e2205326119. Retrieved from https://doi.org/10.1073/pnas.2205326119 (doi: 10.1073/pnas.2205326119) doi: 10.1073/pnas.2205326119

Trommer, G., Siccha, M., van der Meer, M. T. J., Schouten, S., Sinninghe Damsté, J. S., Schulz, H., ... Kucera, M. (2009). Distribution of crenarchaeota tetraether membrane lipids in surface sediments from the red sea. *Organic Geochemistry*, *40*, 724-731. Retrieved from https://www.sciencedirect.com/science/article/pii/S0146638009000631 doi: https://doi.org/10.1016/j.orggeochem.2009.03.001

Uda, I., Sugai, A., Itoh, Y. H., & Itoh, T. (2001). Variation in molecular species of polar lipids from Thermoplasma acidophilum depends on growth temperature. *Lipids*, *36*(1), 103–105.

Valentine, D. L. (2007). Adaptations to energy stress dictate the ecology and evolution of the archaea. *Nature Reviews Microbiology*, *5*, 316-323.

van der Weijst, C. M. H., van der Laan, K. J., Peterse, F., Reichart, G.-J., Sangiorgi, F., Schouten, S., ... Sluijs, A. (2022, 8). A 15-million-year surface- and subsurface-integrated tex86 temperature record from the eastern equatorial atlantic. *Clim. Past*, *18*, 1947-1962. Retrieved from https://cp.copernicus.org/articles/18/1947/2022/https://cp.copernicus.org/articles/18/1947/2022/cp-18-1947-2022.pdf doi: 10.5194/cp-18-1947-2022

Varma, D., Hopmans, E. C., van Kemenda, Z. R., Kusch, S., Berg, S., Bale, N. J., ... Schouten, S. (2024). *Ohgdgt global surface sediment data.* PANGAEA. Retrieved from https://doi.org/10.1594/PANGAEA.964885 doi: 10.1594/PANGAEA.964885

Varma, D., Villanueva, L., Bale, N. J., Offre, P., Reichart, G.-J., & Schouten, S. (2024). Controls on the composition of hydroxylated isoprenoidal glycerol dialkyl glycerol tetraethers (isogdgts) in cultivated ammonia-oxidizing thaumarchaeota. *Biogeosciences*, *21*, 4875-4888. Retrieved from https://bg.copernicus.org/articles/21/4875/2024/ doi: 10.5194/bg-21-4875-2024

Verleye, T. J. (2011). *The late quaternary palaeoenvironmental changes along the western south-american continental slope: A reconstruction based on dinoflagellate cysts and tex86* (Doctoral dissertation, Ghent University). Retrieved from https://www.vliz.be/en/imis?module=ref&refid=210274

Wei, B., Jia, G., Hefter, J., Kang, M., Park, E., Wang, S., & Mollenhauer, G. (2020). Comparison of the $\mathrm{U}_{37}^{\mathrm{K}'}$, LDI, $\mathrm{TEX}_{86}^{\mathrm{H}}$, and RI-OH temperature proxies in sediments from the northern shelf of the South China Sea. *Biogeosciences*, *17*, 4489-4508. Retrieved from https://bg.copernicus.org/articles/17/4489/2020/ doi: 10.5194/bg-17-4489-2020

Wei, Y., Wang, J., Liu, J., Dong, L., Li, L., Wang, H., ... Zhang, C. L. (2011). Spatial variations in archaeal lipids of surface water and core-top sediments in the south china sea and their implications for paleoclimate studies. *Applied and Environmental Microbiology*, *77*, 7479-7489. Retrieved from https://aem.asm.org/content/aem/77/21/7479.full.pdf doi: 10.1128/aem.00580-11

Windler, G., Tierney, J. E., DiNezio, P. N., Gibson, K., & Thunell, R. (2019). Shelf exposure influence on indo-pacific warm pool climate for the last 450,000 years. *Earth and Planetary Science Letters*, *516*, 66-76. Retrieved from https://www.sciencedirect.com/science/article/pii/S0012821X19301931 doi: https://doi.org/10.1016/j.epsl.2019.03.038

Wu, W., Tan, W., Zhou, L., Yang, H., & Xu, Y. (2012). Sea surface temperature variability in southern okinawa trough during last 2700 years. *Geophysical Research Letters*, *39*. Retrieved from https://doi.org/10.1029/2012GL052749 doi: https://doi.org/10.1029/2012GL052749

Wuchter, C., Schouten, S., Coolen, M. J. L., & Sinninghe Damsté, J. S. (2004). Temperature-dependent variation in the distribution of tetraether membrane lipids of marine Crenarchaeota: Implications for TEX86 paleothermometry. *Paleoceanography*, *19*(4). Retrieved from https://doi.org/10.1029/2004PA001041 doi: https://doi.org/10.1029/2004PA001041

Yamamoto, M., Sai, H., Chen, M.-T., & Zhao, M. (2013). The east asian winter monsoon variability in response to precession during the past 150 000 yr. *Climate of the Past*, *9*, 2777-2788. Retrieved from https://cp.copernicus.org/articles/9/2777/2013/ doi: 10.5194/cp-9-2777-2013

Yang, Y., Gao, C., Dang, X., Ruan, X., Lü, X., Xie, S., ... Yang, H. (2018). Assessing hydroxylated isoprenoid gdgts as a paleothermometer for the tropical south china sea. *Organic Geochemistry*, *115*, 156-165. Retrieved from https://

www.sciencedirect.com/science/article/pii/S0146638017301687 doi: https://doi.org/10.1016/j.orggeochem.2017.10.014

Zachos, J. C., Schouten, S., Bohaty, S., Quattlebaum, T., Sluijs, A., Brinkhuis, H., ... Bralower, T. J. (2006). Extreme warming of mid-latitude coastal ocean during the paleocene-eocene thermal maximum: Inferences from tex86 and isotope data. *Geology*, *34*, 737-740. Retrieved from https://doi.org/10.1130/G22522.1 doi: 10.1130/g22522.1

Zell, C., Kim, J.-H., Hollander, D., Lorenzoni, L., Baker, P., Silva, C. G., ... Sinninghe Damsté, J. S. (2014). Sources and distributions of branched and isoprenoid tetraether lipids on the amazon shelf and fan: Implications for the use of gdgt-based proxies in marine sediments. *Geochimica et Cosmochimica Acta*, *139*, 293-312. Retrieved from http://www.sciencedirect.com/science/article/pii/S0016703714002968 doi: https://doi.org/10.1016/j.gca.2014.04.038

Zhang, Y. G., Pagani, M., & Wang, Z. (2016). Ring Index: A new strategy to evaluate the integrity of TEX86 paleothermometry. *Paleoceanography*, *31*(2), 220–232.

Zhang, Y. G., Zhang, C. L., Liu, X.-L., Li, L., Hinrichs, K.-U., & Noakes, J. E. (2011). Methane index: a tetraether archaeal lipid biomarker indicator for detecting the instability of marine gas hydrates. *Earth and Planetary Science Letters*, *307*, 525-534.

Zhou, H., Hu, J., Spiro, B., Peng, P., & Tang, J. (2014). Glycerol dialkyl glycerol tetraethers in surficial coastal and open marine sediments around china: Indicators of sea surface temperature and effects of their sources. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *395*, 114-121. Retrieved from https://www.sciencedirect.com/science/article/pii/S0031018213005361 doi: https://doi.org/10.1016/j.palaeo.2013.12.006

Zhu, J., Poulsen, C. J., & Tierney, J. E. (2019). Simulation of eocene extreme warmth and high climate sensitivity through cloud feedbacks. *Science Advances*, *5*, eaax1874. Retrieved from https://advances.sciencemag.org/content/advances/5/9/eaax1874.full.pdf doi: 10.1126/sciadv.aax1874